MASTER IN ARTIFICIAL INTELLIGENCE

PLANNING AND APPROXIMATE REASONING

Autonomous Drone Planning for Smart-City Emergency Response

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1. Introduction

Drones can be useful during emergencies, especially when roads are blocked by floods, debris, or damaged infrastructure. They can fly over these obstacles to inspect affected areas and deliver essential supplies such as medical kits, water, and communication equipment. However, coordinating multiple drones is not simple. Each drone has limited battery capacity, can carry only one supply at a time, and may need to visit certain locations before others become accessible.

In this project, we use PDDL (Planning Domain Definition Language) and the Fast Downward planner to plan drone movements, inspections, and supply deliveries. We develop two different ways of modelling battery recharging and test them across four problem instances of increasing size. We use A* and LAMA-first to compare the plans and measure how much effort the planners require. Finally, we validate the generated plans using VAL and discuss the results, limitations, and possible improvements.

2. Problem Analysis

2.1. State Space Representation and Combinatorial Complexity

In classical STRIPS/ADL planning, a planning task is represented by the tuple:

$$\Pi = \langle F, A, s_0, G, c \rangle$$

where F is the set of ground atoms, A the set of ground actions, s_0 the initial state, G the goal conditions, and $(c: A \rightarrow \mathbb{R}_\{\geq 0\})$ the action cost function (Ghallab et al., 2004).

For our domain, let D be the number of drones, L the number of locations, $B + 1$ the number of discrete battery levels, and N_s the number of supply items. The possible configurations of each state component are:

- Drone positions: L^D, since each drone occupies one location.
- Battery levels: $(B + 1)^D$, since each drone has one of $B + 1$ possible levels.
- Supply status: At most $(D + 2)^{N_s}$, since each supply can be at its depot, carried by one of the drones, or delivered.
- Inspection status: 2^L, since each location can be inspected or uninspected. Accessibility is determined by inspection and delivery dependencies rather than counted independently.

Combining these components gives an upper bound on the reachable state space:

$$|S| \leq L^D . (B + 1)^D . (D + 2)^{N_s} . 2^L$$

This is a theoretical upper bound, as some combinations may be invalid or unreachable. Table 1 shows how these parameters and the corresponding state-space bounds grow across our problem instances.

Instance	Drones	Locations	Directed	Battery	Supplies	Goals	State-space
Small (problem-small)	1	5	8	5 (b_0–b_4)	1 (1 type)	3	2.4×10^3
Medium (problem-medium)	2	8	14	7 (b_0–b_6)	3 (2 types)	7	5.1×10^7
Large, 1 drone (large-1drone)	1	11	20	7 (b_0–b_6)	4 (3 types)	7	1.3×10^7
Large, 3 drones (problem-large)	3	11	20	7 (b_0–b_6)	4 (3 types)	7	5.9×10^{11}

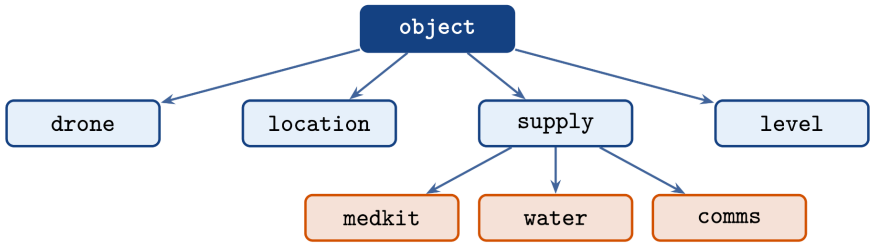
Table 1: The structural parameters and corresponding theoretical state-space bounds for our four problem instances.

2.2. Type Hierarchy Design

Our PDDL domain uses four main types: drone, location, supply, and level. All four are defined as direct subtypes of the base type object. The supply type is further divided into three subtypes representing the emergency cargo categories:

medkit: Medical supplies.
water: Water packages.
comms: Communication equipment.

Figure 1 illustrates the resulting type hierarchy.



2.3 Symbolic Battery Encoding and Arithmetic Tables

Since Fast Downward does not support general numeric state fluents for battery management, we represent battery levels using a finite set of symbolic objects:

$$b_0, b_1, \ldots, b_B \in level$$

Battery consumption and recharging are handled through four static predicates defined in the problem's :init section:

1. (flight-cost ?from ?to - location ?c - level): Associates each directed edge (u, v) with its energy cost c, represented by a discrete battery level (e.g., b_1 or b_2).
2. (battery-after ?current ?c ?left - level): Represents the subtraction relation:
   current-c=left
   Facts are defined only when $current \geq c$. For example, no fact exists for (battery-after b1 b2 ?left). Therefore, a drone with battery b_1 cannot perform a flight costing b_2, preventing battery levels from becoming negative.
3. (next-level ?n1 ?n2 - level): Represents the successor relation:
   n_2=n_1+1
   It is used by Model B to implement incremental recharging.
4. (max-level ?full - level): Identifies the maximum battery level (e.g., b_4 in Small and b_6 in Medium and Large). It is used by Model A to restore the battery to full capacity.

2.3.1 Mission Cost Representation

While battery feasibility is handled symbolically, plan costs are maintained numerically using the :action-costs requirement and the (total-cost) function:

- Every fly action increases (total-cost) by (distance ?from ?to), matching the corresponding flight energy cost.
- Every non-flight action (inspect, pick-up-supply, deliver-supply, and recharge) adds 1.
- Each problem initializes (total-cost) to 0 and specifies (:metric minimize (total-cost)).

This separates battery feasibility from cost optimization: symbolic predicates determine whether a flight is possible, while numeric action costs determine the cheapest valid plan (Helmert, 2006).

2.4 Key Design Decisions

Explicit (empty ?d) Predicate
To enforce the single-item payload restriction, we use an explicit dynamic predicate (empty ?d) instead of checking that a drone is not carrying any supply through a universally quantified negative condition. The pick-up-supply action requires (empty ?d), deletes it upon pickup, and adds (carrying ?d ?s). The deliver-supply action reverses this transition.

This simplifies the action preconditions and allows Fast Downward to represent the mutually exclusive payload states as a single finite-domain variable with $N_s + 1$ possible values per drone (Helmert, 2006, 2009).

2.4.1 Universal Conditional Effects for Dynamic Unlocking
Locations that become accessible after an inspection or delivery are handled through two static dependency predicates:

- (unlocks-by-inspection ?l ?y): Inspecting location ?l unlocks ?y.
- (unlocks-by-delivery ?s ?l ?y): Delivering supply ?s at ?l unlocks ?y.

The inspect and deliver-supply actions use universally quantified conditional effects:

```
(forall (?y - location)
  (when (unlocks-by-inspection ?l ?y)
    (accessible ?y)))
```

This allows one general action schema to unlock all locations matching the defined dependencies, avoiding separate unlocking actions or additional target-location parameters (McDermott et al., 1998).

3. PDDL Implementation

3.1 Predicates

Both Domain A and Domain B have 17 predicates. These predicates are divided into two: *dynamic* where it is changed during execution of the plan and *static* where it is fixed in the initial state. Table 2 provides a detailed reference of every predicate, mutability, and role .The full code block on how we wrote these is seen at Appendix A1.

Predicate	Static or Dynamic	Description
DRONE		
(at ?d - drone ?l - location)	Dynamic	Drone ?d is at a location ?l. Updated by fly.
(carrying ?d - drone ?s - supply)	Dynamic	Drone ?d is carrying ?s. Added by pick-up-supply, deleted by deliver-supply
(empty ?d - drone)	Dynamic	Drone ?d is carrying no supply. Deleted by pick-up-supply.
(battery ?d - drone ?current - level)	Dynamic	Drone ?d has battery level ?current. Updated by fly and recharge.
MAP		
(connected ?l1 ?l2 - location)	Static	Drone can fly between ?l1 to ?l2
(has-charger ?l - location)	Static	Location ?l has a charging station. Required precondition for recharge.
(accessible ?l - location)	Dynamic	Location ?l is accessible. Checked by fly, added by inspect or deliver-supply.
SUPPLIES AND MISSION		
(supply-at ?s - supply ?l - location)	Dynamic	Supply ?s is available at location ?l. Deleted upon pickup
(delivered ?s - supply ?l - location)	Dynamic	Supply ?s is delivered to location ?l. Used in :goal conditions
(inspected ?l - location)	Dynamic	Location ?l is inspected by drone. Used in :goal conditions and negative precondition of inspect.
(needs ?s - supply ?l - location)	Dynamic	Location ?l requires supply ?s. Checked and cleared by deliver-supply.
UNLOCKING RULES		
(unlocks-by-inspection ?l ?y - location	Static	Inspecting location ?l makes locked location ?y accessible.
(unlocks-by-delivery ?s - supply ?l ?y - location)	Static	Delivering supply ?s to location ?l makes location ?y accessible.
BATTERY ENCODING		
(flight-cost ?from ?to - location ?c - level)	Static	Energy cost ?c required to fly along edge ?from ?to
(battery-after ?current ?c ?left - level?	Static	Reference table for ?current - ?c = ?left. Only applies ?current ?c to prevent negative battery states.
(next-level ?n1 ?n2 - level)	Static	?n2 = ?n1 + 1. Used by recharge in Model B.
(max-level ?full - level)	Static	Maximum battery ?full. Used by recharge in Model A.

Table 2: Complete specification of predicates in domain-A.pddl and domain-B.pddl

3.2 Action Schemas

There are five action schemas implemented. The first four actions (1) fly, (2) inspect, (3) pick-up supply, (4) deliver-supply are identical in Model A and Model B. The fifth action (5) recharge implements the model's specific battery charging rule. Table 3 illustrates the domain actions, its parameters, preconditions, effects, and costs. The full code for the action schemas can be seen in Appendix A2.

Action	Parameters	Preconditions	Effects (+ add, - delete)	Cost	Comment
fly	?d ?from ?to ?current ?c ?left	(at ?d ?from), (connected ?from ?to), (accessible ?to), (battery ?d ?current), (flight-cost ?from ?to ?c), (battery-after ?current ?c ?left)	(not (at ?d ?from)), (not (battery ?d ?current)), (at ?d ?to), (battery ?d ?left)	distance	Battery use and plan cost are separate. Battery use (?c) limits feasibility, while distance drives the optimisation
inspect	?d ?l	(at ?d ?l), (inspected ?l)	(inspected ?l); ∀?y: (unlocks-by-inspection ?l ?y) → (accessible ?y)	1	The conditional effect models dependencies between areas: inspecting one area opens access to others
pick-up-supply	?d ?s ?l	(at ?d ?l), (supply-at ?s ?l), (empty ?d)	(not (supply-at ?s ?l)), (not (empty ?d)), (carrying ?d ?s)	1	(empty ?d) limits each drone to one supply at a time
deliver-supply	?d ?s ?l	(at ?d ?l), (carrying ?d ?s), (needs ?s ?l)	(not (carrying ?d ?s)), (not (needs ?s ?l)), (empty ?d), (delivered ?s ?l); ∀?y: (unlocks-by-delivery ?l ?y) → $^{+}$(accessible ?y)	1	(needs ?s ?l) ensures supplies are delivered only to their intended recipient
recharge (Model A)	?d ?l ?current ?full	(at ?d ?l), (has-charger ?l), (battery ?d ?current), (max-level ?full)	(not (battery ?d ?current)), (battery ?d ?full)	1	One action fully recharges the drone, which gives shorter plans
recharge (Model B)	?d ?l ?current ?next	(at ?d ?l), (has-charger ?l), (battery ?d ?current), (next ?current ?next)	(not (battery ?d ?current), (battery ?d ?next)	1	Each action adds one battery level, so the planner chooses how much to charge. Plans are longer but charging is more precise

Table 3. Domain actions: parameters, preconditions, effects and cost

4. Alternative Models

Both models use the same predicates, battery levels, and core actions. Their main difference is how the recharge action works. Model A restores the battery directly to its maximum level, while Model B increases it one level at a time. Figure 2 illustrates these two approaches.

4.1 Expressiveness

The main difference between the models is how realistically they represent battery charging.

- Model A – Full Recharge: A single recharge action restores the battery to its maximum level, regardless of its current charge. For example, recharging from b_0 to b_6 costs the same as recharging from b_5 to b_6: one action with cost 1. This represents an idealized battery-swapping station but does not support partial charging.

- Model B – Incremental Recharge: Each recharge action increases the battery by one level using the next-level predicate. Gaining Δb levels therefore requires Δb actions, with a total cost of Δb. This allows drones to recharge only as much as needed for their remaining tasks.

For example, in the optimal Small and Medium plans under Model B, droneA arrives at a charging station with b_1 and performs only two recharge actions, reaching b_3 instead of charging to full capacity.

4.2 Planning Complexity

Model A is generally easier for both A* and LAMA-first to solve because it uses fewer recharge actions and creates shorter plans.

In Model B, a recharge that takes one action in Model A may require several consecutive actions. This increases plan length and creates more intermediate states for the planner to explore. In multi-drone problems, these intermediate states also allow actions from different drones to be interleaved in more possible ways.

The effect is visible in our experiments:

- In the Large 1-Drone instance, A* expanded 1,107 states in Model A compared to 3,331 in Model B (approximately 3 times more).
- Under LAMA-first, expansions increased from 99 to 9,304 in the Medium instance (94 times more), and from 49 to 9,566 in the Large 3-Drone instance (195 times more).

Thus, Model B provides more realistic charging behaviour but can make the search considerably harder.

4.4 Summary of Trade-Offs

Dimension	Model A (domain-A.pddl)	Model B (domain-B.pddl)
Recharge behaviour	Restores battery to full in one action.	Increases battery by one level per action.
Realism	Represents idealized battery swapping.	Represents gradual charging more realistically.
Partial charging	Not supported.	Supported; drones can charge only as much as needed.
Planning complexity	Shorter plans and generally fewer expanded states.	Longer plans and potentially much larger search spaces.
Plan quality	Cheap full recharging can encourage unnecessary battery use and poor task allocation.	Recharge costs better reflect energy replenishment and can encourage task sharing.
Main advantage	Faster and simpler planning.	More realistic charging and greater flexibility.
Main limitation	Less realistic charging costs and no partial recharging.	Greater search effort and longer plans.

Table 4: Comparison of Model A and Model B.

5. Problem Instances

There are three problem instances in this project: small, medium, and large. Large has an additional variant that runs on only 1 drone.

5.1 Small Instance

Fleet & Battery Capacity: 1 drone (dronaA) starts at depot with full battery (b4); 5 discrete levels (b0-b4, capacity 4); and all flight edge costs equal to 1 level which uses 1 battery level.
Locations (5): 1 depot (depot), 1 charging station (charging), and 3 mission sites (s1, s2, s3) are arranged in a linear chain s3 ↔ charging ↔ depot ↔ s1 ↔ s2. All locations are initially accessible.
Supplies and Goals: Supply kit 1 (medkit) starts at depot and is needed at s1. Three goals are (inspected s2), (inspected s3), and (delivered kit1 s1).

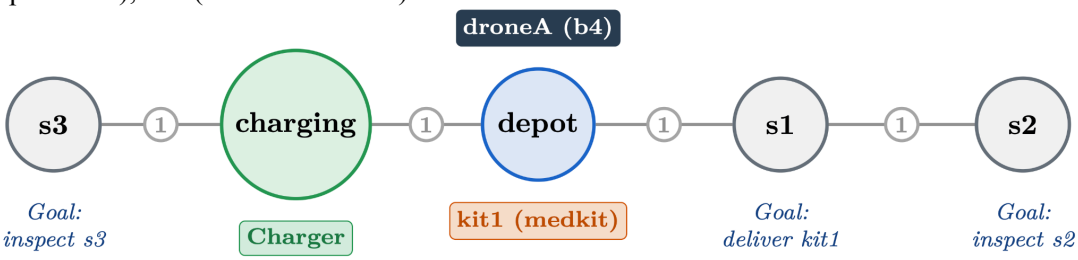


*Figure 2. Location diagram and initial state for the **Small instance**. Battery capacity has 4 levels (b0-b4)*

It is true that the drone must recharge at least once to complete the mission. In the figure graph above, droneA starts at the central vertex depot with battery capacity of 4. Every plan must visit both extremes leaves s3 ($d(depot, s3) = 2$ to the left) and s2 ($d(depot, s2) = 2$ to the right, passing through s1 where kit 1 is delivered). Any flight starting at the depot that visits both s3 and s2 must go through one side completely, costing 2 battery levels, and returning to depot costs another 2 levels, and going to the other side costs another 2. This can be proven mathematically:

$$Minimum\ flight\ energy = min(2 \cdot d(depot, s3) + d(depot, s2),\ 2 \cdot d(depot, s2) + d(depot, s3)) \\ = 2(2) + 2 = 6\ battery\ levels$$

Since the required energy of 6 levels exceeds the battery capacity of 4 levels, any path without a recharge depletes the battery to b0 after 4 flights. As a result, the drone either gets stranded at the depot upon returning from the first branch, or ends up 2 steps short of the second destination. Therefore, at least one recharge at charging is mandatory.

5.2 Medium Instance

Fleet & Battery Capacity: 2 drones (dronaA at depot with b6, droneB at charging2 with b6); 7 discrete battery levels (b0-b6, capacity 6); flight edge costs vary between 1 (b1) and 2 (b2).
Locations (8): 1 depot (depot), 2 charging stations (charging1 and charging2), and 5 mission sites (s1, s2, s3, s4, s5). All locations are accessible except s5, which is only accessible by inspecting s4.
Supplies and Goals: There are 3 supplies: 2 medical kits (medkit1 and medkit 2) and 1 water package (water1). There are 7 goals: inspect 4 locations (s2, s3, s4, and s5), and deliver 3 supplies (medkit1 to s1, medkit2 to s2, and water1 to s5).

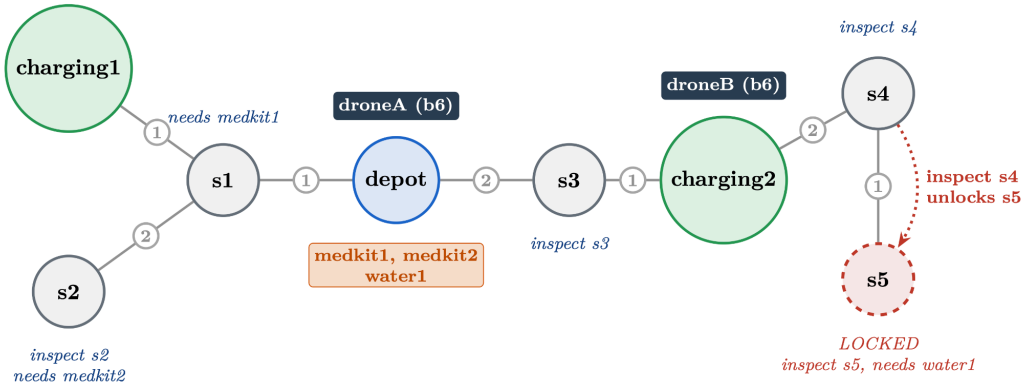


*Figure 3. Location diagram and initial state for the **Medium instance**. Battery capacity has 6 levels (b0-b6). Site s5 is locked until s4 is inspected.*

If droneB starts at charging2 with battery b6 and tries to reach s5 without inspect s4 first:

1. (fly droneB charging2 s4 b6 b2 b4)
   (at droneB charging2), (connected charging2 s4), (accessible s4), (flight-cost charging2 s4 b2), and (battery-after b6 b2 b4) all satisfy the preconditions in the initial state. The state updates to (at droneB s4) and (battery droneB b4).
2. (fly droneB s4 s5 b4 b1 b3)
   This action is blocked. droneB is at s4, (connected s4 s5) is true, and droneB has enough battery (battery-after b4 b1 b3). However, the third precondition of fly requires (accessible s5), which is false in the initial state. (accessible s5) can only become true through the conditional effect of (inspect ?d s4): because (unlocks-by-inspection s4 s5) holds, inspecting s4 adds (accessible s5). Therefore, (fly droneB s4 s5 b4 b1 b3) is not applicable until (inspect ?d s4) has been executed.

5.3 Large Instance (3-Drone and 1-Drone Variants)

Fleet & Battery Capacity: For the 3-drone variant (dronaA at depot1, droneB at charging2, droneC at depot2, all starting with b6); 1 drone (droneA at depot1 with b6). There are 7 discrete battery levels (b0-b6, capacity 6); flight edge costs vary between 1 (b1) and 2 (b2).

Locations (11): 2 depots (depot1, depot2), 3 charging stations (charging1, charging2, charging 3), and 6 mission sites (s1, s2, s3, s4, s5, s6). All locations are accessible except s4, which is only accessible by picking up comms1 at depot2 and delivering it to s3.

Supplies and Goals: There are 3 supplies: 2 medical kits (medkit1 and medkit 2) and 1 water package (water1). There are 7 goals: inspect 4 locations (s2, s3, s4, and s5), and deliver 3 supplies (medkit1 to s1, medkit2 to s2, and water1 to s5).

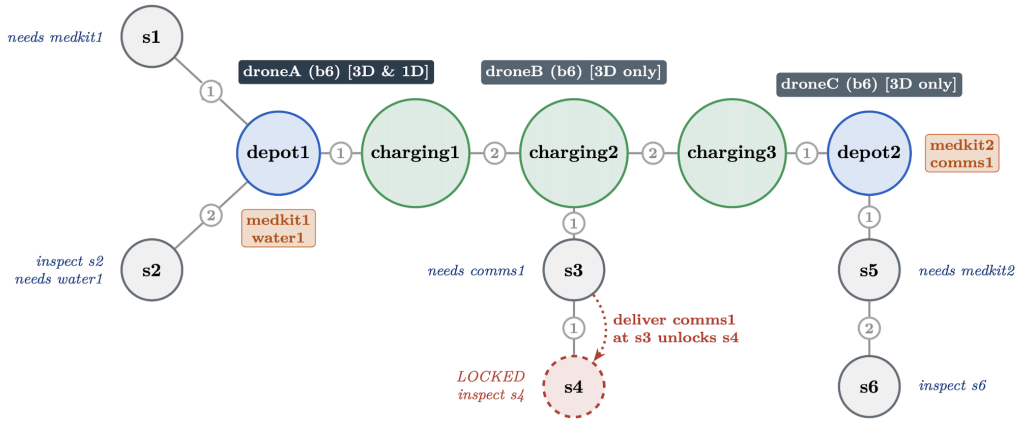


*Figure 4. Location diagram and initial state for the **Large and Large 1-Drone instances**. Battery capacity has 6 levels (b0-b6). Site s4 is locked until comms1 is delivered at s3.*

5.4 1-Drone vs 3-Drone Comparison

The large instance evaluates how deploying a three-drone fleet reduces cost compared to a single drone executing the same mission. Table 4 presents the side-by-side comparison of all validated plans across both domain models and both planner configurations

Domain	Planner	1-Drone Cost	3-Drone Cost	Abs. Saving	% Cost Reduction	Recharges	
						1D	3D
Model A	A*	42 (35 steps)	26 (22 steps)	-16	-38.10%	4	0
Model B		59 (52 steps)	26 (22 steps)	-33	–55.93%	21	0
Model A	Lama-First	56 (47 steps)	34 (28 steps)	-22	-39.29%	6	2
Model B		95 (84 steps)	36 (30 steps)	-59	-62.11%	45	4

Table 4. Comparison of 1-Drone vs 3-Drone variants on the Large instance

Under Model A, 3 drones reduce the optimal cost from 42 to 26, which saves 38.10% in costs. On the other hand under Model B, 3 drones reduce the cost from 59 to 26, which saves 55.93%. This happens as 3 drones completely eliminates the need for recharging, allowing the drones to achieve the same cost of 26 in both models. Model B gets a larger reduction because a single drone that does +1 step charging only requires more trips to the charging station.

6.Planner Setup

6.1 Tools and Environment

All experiments were done locally on macOS with the following software and hardware:

- **Planner: Fast Downward**, compiled natively from C++20 source on macOS.

- **Plan Validator: VAL (KCL-Planning/VAL)** installed locally from source using CMake (Validate binary, applied with the verbose -v flag).
- **Hardware & OS:** Macbook Pro M5 Max running macOS Golden Gate 27.2 with 36GB unified RAM

6.2 Search Configurations Tested

We evaluated every domain model and problem instance across two Fast Downward search configurations.

1. ***Cost-Optimal Search (-search "astar(lmcut())"***

The goal is to find the cheapest possible, no matter how long the search takes. It uses the A* algorithm ($f(s) = g(s) + h_{LM-cut}(s)$) to evaluate states and by combining the cost already spent (g) with an estimate of the remaining cost to the goal (h). To compute the remaining cost, the LM-Cut heuristic identifies the bottleneck actions, called landmarks, that must occur to reach the goal, and sums their minimum costs. Because the LM-Cut never overestimates the remaining cost, the plan it discovers is the lowest possible cost.

2. ***Satisficing Search (-alias lama-first)***

The goal is to find any working solution as fast as possible, without worrying about plan costs. It is a greedy best-first search method, which rushes towards the finished goal without keeping track of the total cost. It evaluate states using two heuristics at the same time across multiple search queues: the h_{FF} heuristic to estimate how many steps are left, while the h_{LM} landmark-count heuristic tracks how many milestones have been reached. To speed up execution, the search prioritizes helpful actions to avoid wasting time on unpromising moves, and uses lazy evaluation to delay computing complex heuristics until a state is expanded. The trade off to this quick search method is that the costs might be significantly more expensive.

6.3 Experimental Procedure

Every run was applied with a 5-minute timeout (-overall-time-limit 5m). This is due to the limitations of macOS and VAL extension.

macOS Kernel setrlimit Restriction: The Darwin kernel does not support setting virtual memory limits via RLIMIT_AS / ulimit -v. Passing -overall-memory-limit 4G to Fast Downward on macOS causes the Python driver to abort immediately before translation with 'Setting memory limits is not supported on your platform' (exit code 37). Removing that flag allowed all 16 runs to execute cleanly; and every run completed under 0.54 seconds and consumed less than 20 MB of memory.

VAL Extension Handling: The native macOS build of VAL's validate binary rejected plan filenames containing a .plan suffix (Bad plan file!), where extensionless filenames (eg., plans/small-A-astar) were parsed and validated without issue. All 16 generated plans were stored in plans/ and verified with Validate -v, with full outputs saved in validation/.

6.4 AI Usage

AI tools were used to support the development and documentation of the project in the following ways:

- Identifying potential errors and helping troubleshoot issues in the implementation.
- Diagnosing and resolving errors encountered when using the planning and validation tools.
- Improving and expanding comments in the .pddl files to make the domain models easier to understand.
- Checking the report for grammatical errors and improving sentence clarity.
- Shortening longer sections of the report to meet the report's length constraints while preserving the key information.

7 Results and Analysis

7.1 Experimental Results

Overall, Model A generally requires less search effort, while Model B provides more realistic charging behaviour at the cost of longer plans and increased search complexity.

Table 5: Complete empirical evaluation results across all 16 combinations of problem instance, domain model, and planner configuration. Every reported plan is validated by VAL (Plan valid).

Instance	Domain	Planner	Solved	Search (s)	Total (s)	Length	Cost	Expanded	Evaluated	Generated
Small	Model A	`astar(lmcut())`	Yes	0.0003	0.0013	11	**11**	33	60	68
	Model A	`lama-first`	Yes	0.0002	0.0015	11	**11**	37	54	82
	Model B	`astar(lmcut())`	Yes	0.0003	0.0013	12	**12**	49	83	106
	Model B	`lama-first`	Yes	0.0002	0.0015	16	16	48	71	102
Medium	Model A	`astar(lmcut())`	Yes	0.0869	0.0888	21	**25**	1,959	5,669	9,131
	Model A	`lama-first`	Yes	0.0006	0.0029	38	49	99	109	316
	Model B	`astar(lmcut())`	Yes	0.1667	0.1686	22	**26**	3,904	10,756	18,071
	Model B	`lama-first`	Yes	0.0310	0.0333	23	27	9,304	12,046	32,477
Large (3D)	Model A	`astar(lmcut())`	Yes	0.5143	0.5170	22	**26**	3,806	19,054	28,172
	Model A	`lama-first`	Yes	0.0004	0.0040	28	34	49	50	257
	Model B	`astar(lmcut())`	Yes	0.5327	0.5353	22	**26**	3,834	19,504	28,658
	Model B	`lama-first`	Yes	0.0385	0.0421	30	36	9,566	11,444	40,244
Large (1D)	Model A	`astar(lmcut())`	Yes	0.0184	0.0198	35	**42**	1,107	1,524	2,483
	Model A	`lama-first`	Yes	0.0006	0.0027	47	56	124	145	294
	Model B	`astar(lmcut())`	Yes	0.0512	0.0527	52	**59**	3,331	4,481	7,676
	Model B	`lama-first`	Yes	0.0022	0.0043	84	95	648	891	1,386

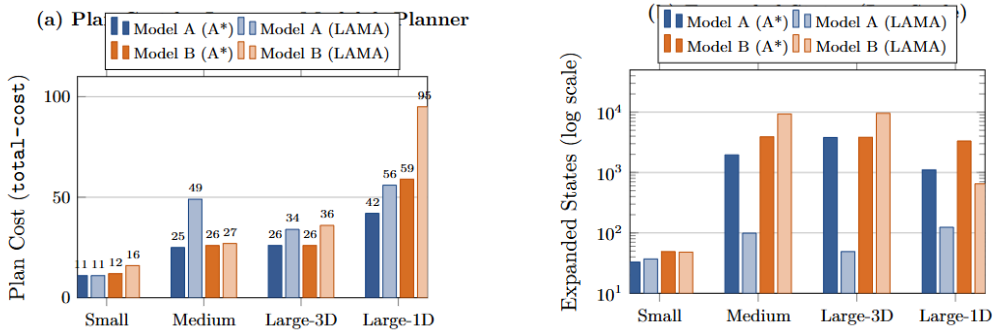


Figure 6: Empirical comparison across all 16 experimental runs: (a) validated plan cost (total-cost) and (b) number of expanded states on a logarithmic scale.

7.2 Analysis of Results

7.2.1 Small Instance

The Small instance was easy for both planners, with all four configurations finishing in approximately 1.3–1.5 milliseconds and expanding fewer than 50 states.

Model A produced a shorter optimal plan (11 actions, cost 11) compared with Model B (12 actions, cost 12). This difference comes from the recharge mechanism: Model A restores the battery in one action, whereas Model B requires two incremental recharge actions. LAMA-first also performed better under Model A, finding an optimal plan, while its Model B solution had a cost of 16.

Overall, the small state space makes both models easy to solve, although Model A remains slightly more efficient.

7.2.2 Medium Instance

The Medium instance introduced greater complexity due to two interacting drones, different flight costs, and the dependency that requires s4 to be inspected before s5 becomes accessible.

Under A*, switching from Model A to Model B nearly doubled the number of expanded states (1,959 to 3,904), while the optimal plan cost increased only slightly (25 to 26).

A more interesting difference appeared under LAMA-first. Model A produced a relatively poor plan with cost 49, while Model B achieved a cost of 27, close to the optimal cost of 25–26. The incremental recharge cost in Model B encouraged the planner to distribute tasks across both drones rather than relying heavily on one drone.

This shows that a more realistic recharge model can sometimes improve the quality of plans found by satisficing search, even when it requires significantly more search effort.

7.2.3 Large Instance (3-Drone and 1-Drone Variants)

The Large instance highlights the difference between using multiple drones and relying on a single drone.

In the 3-drone variant, A* found identical optimal plans under both models: 22 actions with a cost of 26. Since the three drones can complete their assigned tasks without recharging, the choice of recharge model has almost no effect on the optimal solution.

However, LAMA-first expanded only 49 states under Model A compared with 9,566 under Model B. This difference occurs because the greedy search performs unnecessary recharges under Model B, introducing many additional intermediate states.

The 1-drone variant shows a much larger difference. Since one drone must cover the entire map and recharge multiple times, Model B increases the optimal plan cost from 42 to 59 and the plan length from 35 to 52 actions. A* also expands approximately three times more states (1,107 to 3,331).

This comparison highlights two important observations: multiple drones significantly reduce mission cost by distributing tasks across different regions, but their independent actions also increase the search space. Furthermore, the additional recharge actions in Model B become particularly expensive when a single drone must perform the entire mission.

8. Literature Connection

8.1 Symbolic vs. Numeric Resource Encoding in Classical Planning

Fast Downward uses a propositional/SAS+ representation rather than supporting general numeric battery fluents (Helmert, 2006). To address this limitation, we represent battery levels as finite symbolic objects ($b_0, ..., b_6$) and use static predicates such as battery-after and next-level to model battery consumption and recharging.

Scala et al. (2016) discuss approaches for handling numeric resources in classical planning. Our implementation follows the idea of representing bounded resources symbolically, allowing Fast Downward to reason about battery constraints using its classical planning techniques. However, this also introduces additional grounded actions and intermediate states, particularly when recharging is performed incrementally.

Our experiments reflect this trade-off. Under Model B, LAMA-first expanded 94 times more states in the Medium instance and 195 times more in the Large instance compared with Model A. This shows how a

more detailed resource representation can increase search complexity, even when it provides more realistic charging behaviour.

8.2 Multi-UAV Coordination, Symmetry, and Partial-Order Interleaving

Nissim and Brafman (2014) show that centralized planning for multiple agents can become expensive because independent actions from different agents can be executed in many possible orders. Even when agents have separate tasks, a planner must consider different action interleavings when searching for an optimal plan.

We observed this effect when comparing our Large 1-Drone and 3-Drone instances. Using three drones reduced the optimal mission cost from 42 to 26 under Model A (38.1% reduction) and from 59 to 26 under Model B (55.9% reduction). However, A* expanded more states in the three-drone instance (3,806 vs. 1,107 under Model A) and also required more runtime (0.517s vs. 0.020s).

The reason is that the three drones perform largely independent tasks in different regions of the map. Although this reduces the total mission cost, the planner must still consider different orderings of their actions before proving an optimal plan. This illustrates the trade-off between the benefits of using multiple drones and the additional search complexity in centralized multi-agent planning. Related work on multi-UAV mission planning also explores this coordination problem (Ramirez-Atencia et al., 2017).

9. Limitations and Future Work

There are several interesting directions that could be explored to make the drone-planning model more realistic. One is to use temporal planning (PDDL 2.1), allowing multiple drones to perform actions such as flying, delivering supplies, and recharging at the same time. This would also allow us to focus on minimizing the total mission completion time rather than just the combined action cost.

Another direction is to explore how payload weight and charging station availability affect mission planning. Carrying heavier supplies, such as water packages, could increase battery consumption. Similarly, introducing limited charging pads could require drones to plan when and where to recharge, especially when several drones need the same station.

Finally, uncertainty in battery consumption is worth exploring. Wind and weather conditions can make the energy required for a flight difficult to predict. Probabilistic planning approaches such as PPDDL or RDDL could help represent these uncertainties and allow drones to maintain enough battery reserves for unexpected situations.

10. Appendix

A1. Domain Files

A1.1 Domain A

None

```
; Smart-city emergency drones - Model A (recharge to max)
(define (domain drones)
  (:requirements :strips :typing :negative-preconditions
                 :conditional-effects :action-costs)

  (:types
    drone location supply level - object   ; base types
    medkit water comms - supply            ; kinds of supply
  )

  (:predicates
    ;  Drone state (dynamic - changes during actions)
    (at ?d - drone ?l - location)                          ;where drone is located
    (carrying ?d - drone ?s - supply)        ;whether a drone is carrying a supply
    (empty ?d - drone)                        ;whether a drone is carrying nothing
    (battery ?d - drone ?current - level)            ;the battery level of a drone

    ;  Map (static - does not change during actions)
    (connected ?l1 ?l2 - location)            ;whether two locations are connected
    (has-charger ?l - location)         ;whether a location has a charging station

    ;  Map (dynamic)
    (accessible ?l - location)                   ;whether a location is accessible

    ;  Supplies & Mission (dynamic)
    (supply-at ?s - supply ?l - location)     ;whether a supply is available at a
location
    (delivered ?s - supply ?l - location)      ;whether a supply has been
delivered to a location
    (inspected ?l - location)              ;whether a location has been inspected
    (needs ?s - supply ?l - location)         ;whether a location needs a supply

    ;  Unlocking rules (static)
    (unlocks-by-inspection ?l  ?y - location)   ;inspecting ?l unlocks access to y
    (unlocks-by-delivery ?s - supply ?l ?y - location)    ;delivering ?s at ?l
unlocks access to y

    ;  Battery encoding (static)
    (flight-cost ?from ?to - location ?c - level) ; flying ?from -> ?to costs ?c
levels
    (battery-after ?current ?c ?left - level) ; ?current - ?c = ?left (only listed
if ?current >= ?c)
    (next-level ?n1 ?n2 - level)      ; ?l2 = ?l1 + 1 (used by Model B's recharge)
    (max-level ?full - level)                 ; ?full is the maximum battery level
  )

  (:functions
```

```
    (total-cost) - number                    ; plan cost, minimized by the metric
    (distance ?from ?to - location) - number    ; static: cost added to total-cost
per flight
  )

  (:action fly
    :parameters (?d - drone ?from ?to - location ?current ?c ?left - level)
    :precondition (and
      (at ?d ?from)
      (connected ?from ?to)
      (accessible ?to)
      (battery ?d ?current)
      (flight-cost ?from ?to ?c)
      (battery-after ?current ?c ?left)
    )
    :effect (and
      (not (at ?d ?from))                          ; drone no longer at ?from
      (at ?d ?to)                                  ; drone now at ?to
      (not (battery ?d ?current))             ; drone's battery no longer ?current
      (battery ?d ?left)                           ; drone's battery now ?left
      (increase (total-cost) (distance ?from ?to))
    )
  )

  (:action inspect
    :parameters (?d - drone ?l - location)
    :precondition (and
      (at ?d ?l)                                     ; drone ?d is at ?l
      (not (inspected ?l))                           ; ?l has not been inspected
    )
    :effect (and
      (inspected ?l)                                 ; ?l is now inspected
      (forall (?y - location)
        (when (unlocks-by-inspection ?l ?y)          ; inspecting ?l unlocks ?y
              (accessible ?y))); ?y becomes accessible
      (increase (total-cost) 1)
    )
  )

  (:action pick-up-supply
    :parameters (?d - drone ?s - supply ?l - location)
    :precondition (and
      (at ?d ?l)                                     ; drone ?d is at ?l
      (supply-at ?s ?l)                              ; supply ?s is at ?l
      (empty ?d)                                     ; drone ?d is empty
    )
    :effect (and
      (not (supply-at ?s ?l))                        ; supply ?s is no longer at
?l
      (not (empty ?d)); drone ?d is no longer empty
      (carrying ?d ?s); drone ?d is carrying ?s
```

```
      (increase (total-cost) 1)
    )
  )

  (:action deliver-supply
    :parameters (?d - drone ?s - supply ?l - location)
    :precondition (and
      (at ?d ?l); drone ?d is at ?l
      (carrying ?d ?s); drone ?d is carrying ?s
      (needs ?s ?l); location ?l needs ?s
    )
    :effect (and
      (not (carrying ?d ?s)); drone ?d is no longer carrying ?s
      (empty ?d); drone ?d is empty again
      (delivered ?s ?l); ?s is delivered at ?l
      (not (needs ?s ?l)); ?l no longer needs ?s
      (forall (?y - location)
        (when (unlocks-by-delivery ?s ?l ?y); delivering ?s at ?l unlocks ?y
              (accessible ?y))); ?y becomes accessible
      (increase (total-cost) 1)
    )
  )

  (:action recharge
    :parameters (?d - drone ?l - location ?current ?full - level)
    :precondition (and
      (at ?d ?l); drone ?d is at ?l
      (has-charger ?l); ?l has a charger
      (battery ?d ?current); drone's battery is ?current
      (max-level ?full); ?full is the max level
    )
    :effect (and
      (not (battery ?d ?current)); battery no longer ?current
      (battery ?d ?full); battery is now ?full
      (increase (total-cost) 1)
    )
  )
)
```

A1.2 Domain B

None

```
; Smart-city emergency drones - Model B (recharge +1 level)
(define (domain drones)
  (:requirements :strips :typing :negative-preconditions
                 :conditional-effects :action-costs)

  (:types
    drone location supply level - object   ; base types
```

```
    medkit water comms - supply            ; kinds of supply
  )

  (:predicates
    ;  Drone state (dynamic - changes during actions)
    (at ?d - drone ?l - location)            ;where drone is located
    (carrying ?d - drone ?s - supply)        ;whether a drone is carrying a supply
    (empty ?d - drone)                      ;whether a drone is carrying nothing
    (battery ?d - drone ?current - level)       ;the battery level of a drone

    ;  Map (static - does not change during actions)
    (connected ?l1 - location ?l2 - location)   ;whether two locations are
connected
    (has-charger ?l - location)         ;whether a location has a charging station

    ;  Map (dynamic)
    (accessible ?l - location)               ;whether a location is accessible

    ;  Supplies & Mission (dynamic)
    (supply-at ?s - supply ?l - location)   ;whether a supply is available at a
location
    (delivered ?s - supply ?l - location)   ;whether a supply has been delivered
to a location
    (inspected ?l - location)               ;whether a location has been inspected
    (needs ?s - supply ?l - location)          ;whether a location needs a supply

    ;  Unlocking rules (static)
    (unlocks-by-inspection ?l  ?y - location)  ;inspecting ?l1 unlocks access to y
    (unlocks-by-delivery ?s - supply ?l ?y - location)   ;delivering ?s at ?l1
unlocks access to y

    ;  Battery encoding (static)
    (flight-cost ?from ?to - location ?c - level)  ; flying ?from -> ?to costs ?c
levels
    (battery-after ?current ?c ?left - level)       ; ?current - ?c = ?left (only
listed if ?current >= ?c)
    (next-level ?n1 ?n2 - level)      ; ?l2 = ?l1 + 1 (used by Model B's recharge)
    (max-level ?full - level)                ; ?full is the maximum battery level
  )

  (:functions
    (total-cost) - number                    ; plan cost, minimized by the metric
    (distance ?from ?to - location) - number   ; static: cost added to total-cost
per flight
  )

  (:action fly
    :parameters (?d - drone ?from ?to - location ?current ?c ?left - level)
    :precondition (and
      (at ?d ?from)              ; 1. drone ?d is at ?from
      (connected ?from ?to)      ; 2. ?from and ?to are connected
```

```
      (accessible ?to)            ; 3. ?to is accessible
      (battery ?d ?current)        ; 4. drone's battery is ?current
      (flight-cost ?from ?to ?c)  ; 5. flight ?from -> ?to costs ?c levels
      (battery-after ?current ?c ?left)  ; 6. ?current - ?c = ?left exists (enough
battery)
    )
    :effect (and
      (not (at ?d ?from))                    ; drone no longer at ?from
      (at ?d ?to)                            ; drone now at ?to
      (not (battery ?d ?current))            ; drone's battery no longer ?current
      (battery ?d ?left)                     ; drone's battery now ?left
      (increase (total-cost) (distance ?from ?to))
    )
  )

  (:action inspect
    :parameters (?d - drone ?l - location)
    :precondition (and
      (at ?d ?l) ; drone ?d is at ?l
      (not (inspected ?l))                              ; ?l has not been
inspected
    )
    :effect (and
      (inspected ?l) ; ?l is now inspected
      (forall (?y - location)
        (when (unlocks-by-inspection ?l ?y); inspecting ?l unlocks ?y
              (accessible ?y))); ?y becomes accessible
      (increase (total-cost) 1)
    )
  )

  (:action pick-up-supply
    :parameters (?d - drone ?s - supply ?l - location)
    :precondition (and
      (at ?d ?l); drone ?d is at ?l
      (supply-at ?s ?l); supply ?s is at ?l
      (empty ?d); drone ?d is empty
    )
    :effect (and
      (not (supply-at ?s ?l)); supply ?s is no longer at ?l
      (not (empty ?d)); drone ?d is no longer empty
      (carrying ?d ?s); drone ?d is carrying ?s
      (increase (total-cost) 1)
    )
  )

  (:action deliver-supply
    :parameters (?d - drone ?s - supply ?l - location)
    :precondition (and
      (at ?d ?l); drone ?d is at ?l
      (carrying ?d ?s); drone ?d is carrying ?s
```

```
      (needs ?s ?l); location ?l needs ?s
    )
    :effect (and
      (not (carrying ?d ?s)); drone ?d is no longer carrying ?s
      (empty ?d); drone ?d is empty again
      (delivered ?s ?l); ?s is delivered at ?l
      (not (needs ?s ?l)); ?l no longer needs ?s
      (forall (?y - location)
        (when (unlocks-by-delivery ?s ?l ?y); delivering ?s at ?l unlocks ?y
              (accessible ?y))); ?y becomes accessible
      (increase (total-cost) 1)
    )
  )

  (:action recharge
    :parameters (?d - drone ?l - location ?current ?next - level)
    :precondition (and
      (at ?d ?l); drone ?d is at ?l
      (has-charger ?l); ?l has a charger
      (battery ?d ?current); drone's battery is ?current
      (next-level ?current ?next); ?next is the level above ?current
    )
    :effect (and
      (not (battery ?d ?current)); battery no longer ?current
      (battery ?d ?next); battery is now ?next
      (increase (total-cost) 1)
    )
  )
)
```

A2. Problem Files

A2.1 Problem-Small

```
None
(define (problem small)
  (:domain drones)

  (:objects
    droneA - drone
    s2 s1 depot charging s3 - location
    kit1 - medkit
    b0 b1 b2 b3 b4 - level
  )

  (:init
    ;  Battery tables
    (max-level b4)
    (next-level b0 b1)
    (next-level b1 b2)
```

```
  (next-level b2 b3)
  (next-level b3 b4)

  (battery-after b1 b1 b0)
  (battery-after b2 b1 b1)
  (battery-after b3 b1 b2)
  (battery-after b4 b1 b3)


  ;  Drone
  (at droneA depot)  (battery droneA b4)  (empty droneA)

  ;  Accessibility & unlocking
  (accessible s1)
  (accessible s2)
  (accessible s3)
  (accessible depot)
  (accessible charging)

  ;  Charger & supplies
  (has-charger charging)
  (supply-at kit1 depot)
  (needs kit1 s1)

  ;  Map: each edge needs 6 facts (both directions)
  (connected depot s1) (connected s1 depot)
  (flight-cost s1 depot b1) (flight-cost depot s1 b1)
  (= (distance s1 depot) 1) (= (distance depot s1) 1)

  (connected depot charging) (connected charging depot)
  (flight-cost charging depot b1) (flight-cost depot charging b1)
  (= (distance charging depot) 1) (= (distance depot charging) 1)

  (connected charging s3) (connected s3 charging)
  (flight-cost charging s3 b1) (flight-cost s3 charging b1)
  (= (distance charging s3) 1) (= (distance s3 charging) 1)

  (connected s1 s2) (connected s2 s1)
  (flight-cost s1 s2 b1) (flight-cost s2 s1 b1)
  (= (distance s1 s2) 1) (= (distance s2 s1) 1)

  ; Plan cost
  (= (total-cost) 0)
)

(:goal (and
  (inspected s2)
  (inspected s3)
  (delivered kit1 s1)
))
```

```
  (:metric minimize (total-cost))
)
```

A2.2 Problem-Medium

```
None
; Medium instance: 2 drones, 8 locations, capacity b6, edge costs 1-2
; Design constraint: s5 is locked until s4 is inspected
; Expected failure: (fly ... s4 s5 ...) before (inspect ... s4) violates
(accessible s5)

(define (problem medium)
  (:domain drones)

  (:objects
    droneA droneB - drone
    s1 s2 s3 s4 s5 charging1 charging2 depot - location
    medkit1 medkit2 - medkit
    water1 - water
    b0 b1 b2 b3 b4 b5 b6 - level
  )

  (:init
    ;  Battery tables
    (max-level b6)
    (next-level b0 b1)
    (next-level b1 b2)
    (next-level b2 b3)
    (next-level b3 b4)
    (next-level b4 b5)
    (next-level b5 b6)

    (battery-after b1 b1 b0)
    (battery-after b2 b1 b1)
    (battery-after b3 b1 b2)
    (battery-after b4 b1 b3)
    (battery-after b5 b1 b4)
    (battery-after b6 b1 b5)

    (battery-after b2 b2 b0)
    (battery-after b3 b2 b1)
    (battery-after b4 b2 b2)
    (battery-after b5 b2 b3)
    (battery-after b6 b2 b4)

    ;  Drone
    (at droneA depot)  (battery droneA b6)  (empty droneA)
    (at droneB charging2)  (battery droneB b6)  (empty droneB)

    ;  Accessibility & unlocking
```

```
  (accessible s1)
  (accessible s2)
  (accessible s3)
  (accessible s4)
  (accessible depot)
  (accessible charging1)
  (accessible charging2)
  (unlocks-by-inspection s4 s5)

  ;  Charger & supplies
  (has-charger charging1)
  (has-charger charging2)
  (supply-at medkit1 depot)
  (supply-at medkit2 depot)
  (supply-at water1 depot)
  (needs medkit1 s1)
  (needs medkit2 s2)
  (needs water1 s5)

  ;  Map: each edge needs 6 facts (both directions)
  (connected depot s1) (connected s1 depot)
  (flight-cost s1 depot b1) (flight-cost depot s1 b1)
  (= (distance s1 depot) 1) (= (distance depot s1) 1)

  (connected s1 charging1) (connected charging1 s1)
  (flight-cost charging1 s1 b1) (flight-cost s1 charging1 b1)
  (= (distance charging1 s1) 1) (= (distance s1 charging1) 1)

  (connected s1 s2) (connected s2 s1)
  (flight-cost s1 s2 b2) (flight-cost s2 s1 b2)
  (= (distance s1 s2) 2) (= (distance s2 s1) 2)

  (connected depot s3) (connected s3 depot)
  (flight-cost s3 depot b2) (flight-cost depot s3 b2)
  (= (distance s3 depot) 2) (= (distance depot s3) 2)

  (connected s3 charging2) (connected charging2 s3)
  (flight-cost charging2 s3 b1) (flight-cost s3 charging2 b1)
  (= (distance charging2 s3) 1) (= (distance s3 charging2) 1)

  (connected charging2 s4) (connected s4 charging2)
  (flight-cost s4 charging2 b2) (flight-cost charging2 s4 b2)
  (= (distance s4 charging2) 2) (= (distance charging2 s4) 2)

  (connected s4 s5) (connected s5 s4)
  (flight-cost s4 s5 b1) (flight-cost s5 s4 b1)
  (= (distance s4 s5) 1) (= (distance s5 s4) 1)

  ; Plan cost
  (= (total-cost) 0)
)
```

```
  (:goal (and
    (inspected s2)
    (inspected s3)
    (inspected s4)
    (inspected s5)
    (delivered medkit1 s1)
    (delivered medkit2 s2)
    (delivered water1 s5)
  ))

  (:metric minimize (total-cost))
)
```

A2.3 Problem-Large

None

```
; Large instance: 3 drones, 11 locations, capacity b6, edge costs 1-2
; Design constraint: s4 is locked until comms1 is delivered to s3;
; Expected failure: fly into s4 before comms1 is delivered to s3 violates
(accessible s4)

(define (problem large)
  (:domain drones)

  (:objects
    droneA droneB droneC - drone
    s1 s2 s3 s4 s5 s6 charging1 charging2 charging3 depot1 depot2 - location
    medkit1 medkit2 - medkit
    water1 - water
    comms1 - comms
    b0 b1 b2 b3 b4 b5 b6 - level
  )

  (:init
    ;  Battery tables
    (max-level b6)
    (next-level b0 b1)
    (next-level b1 b2)
    (next-level b2 b3)
    (next-level b3 b4)
    (next-level b4 b5)
    (next-level b5 b6)

    (battery-after b1 b1 b0)
    (battery-after b2 b1 b1)
    (battery-after b3 b1 b2)
    (battery-after b4 b1 b3)
    (battery-after b5 b1 b4)
    (battery-after b6 b1 b5)
```

```
(battery-after b2 b2 b0)
(battery-after b3 b2 b1)
(battery-after b4 b2 b2)
(battery-after b5 b2 b3)
(battery-after b6 b2 b4)

;  Drones
(at droneA depot1)  (battery droneA b6)  (empty droneA)
(at droneB charging2)  (battery droneB b6)  (empty droneB)
(at droneC depot2)  (battery droneC b6)  (empty droneC)

;  Accessibility & unlocking
(accessible s1)
(accessible s2)
(accessible s3)
(accessible s5)
(accessible s6)
(accessible depot1)
(accessible depot2)
(accessible charging1)
(accessible charging2)
(accessible charging3)
(unlocks-by-delivery comms1 s3 s4)

;  Charger & supplies
(has-charger charging1)
(has-charger charging2)
(has-charger charging3)
(supply-at medkit1 depot1)
(supply-at medkit2 depot2)
(supply-at comms1 depot2)
(supply-at water1 depot1)
(needs medkit1 s1)
(needs water1 s2)
(needs comms1 s3)
(needs medkit2 s5)

;  Map: each edge needs 6 facts (both directions)
(connected depot1 s1) (connected s1 depot1)
(flight-cost s1 depot1 b1) (flight-cost depot1 s1 b1)
(= (distance s1 depot1) 1) (= (distance depot1 s1) 1)

(connected depot1 s2) (connected s2 depot1)
(flight-cost s2 depot1 b2) (flight-cost depot1 s2 b2)
(= (distance s2 depot1) 2) (= (distance depot1 s2) 2)

(connected depot1 charging1) (connected charging1 depot1)
(flight-cost charging1 depot1 b1) (flight-cost depot1 charging1 b1)
(= (distance charging1 depot1) 1) (= (distance depot1 charging1) 1)
```

```
    (connected charging1 charging2) (connected charging2 charging1)
    (flight-cost charging2 charging1 b2) (flight-cost charging1 charging2 b2)
    (= (distance charging1 charging2) 2) (= (distance charging2 charging1) 2)

    (connected charging2 s3) (connected s3 charging2)
    (flight-cost s3 charging2 b1) (flight-cost charging2 s3 b1)
    (= (distance charging2 s3) 1) (= (distance s3 charging2) 1)

    (connected s3 s4) (connected s4 s3)
    (flight-cost s4 s3 b1) (flight-cost s3 s4 b1)
    (= (distance s3 s4) 1) (= (distance s4 s3) 1)

    (connected charging2 charging3) (connected charging3 charging2)
    (flight-cost charging3 charging2 b2) (flight-cost charging2 charging3 b2)
    (= (distance charging2 charging3) 2) (= (distance charging3 charging2) 2)

    (connected charging3 depot2) (connected depot2 charging3)
    (flight-cost depot2 charging3 b1) (flight-cost charging3 depot2 b1)
    (= (distance charging3 depot2) 1) (= (distance depot2 charging3) 1)

    (connected depot2 s5) (connected s5 depot2)
    (flight-cost s5 depot2 b1) (flight-cost depot2 s5 b1)
    (= (distance depot2 s5) 1) (= (distance s5 depot2) 1)

    (connected s5 s6) (connected s6 s5)
    (flight-cost s6 s5 b2) (flight-cost s5 s6 b2)
    (= (distance s5 s6) 2) (= (distance s6 s5) 2)

    ; Plan cost
    (= (total-cost) 0)
  )

  (:goal (and
    (inspected s2)
    (inspected s4)
    (inspected s6)
    (delivered medkit1 s1)
    (delivered medkit2 s5)
    (delivered water1 s2)
    (delivered comms1 s3)
  ))

  (:metric minimize (total-cost))
)
```

A2.4 Problem-Large1

```
None
; Large instance: 1 drone, 11 locations, capacity b6, edge costs 1-2
; Design constraint: s4 is locked until comms1 is delivered to s3;
```

```
; Expected failure: fly into s4 before comms1 is delivered to s3 violates
(accessible s4)

(define (problem large-1drone)
  (:domain drones)

  (:objects
    droneA - drone
    s1 s2 s3 s4 s5 s6 charging1 charging2 charging3 depot1 depot2 - location
    medkit1 medkit2 - medkit
    water1 - water
    comms1 - comms
    b0 b1 b2 b3 b4 b5 b6 - level
  )

  (:init
    ;  Battery tables
    (max-level b6)
    (next-level b0 b1)
    (next-level b1 b2)
    (next-level b2 b3)
    (next-level b3 b4)
    (next-level b4 b5)
    (next-level b5 b6)

    (battery-after b1 b1 b0)
    (battery-after b2 b1 b1)
    (battery-after b3 b1 b2)
    (battery-after b4 b1 b3)
    (battery-after b5 b1 b4)
    (battery-after b6 b1 b5)

    (battery-after b2 b2 b0)
    (battery-after b3 b2 b1)
    (battery-after b4 b2 b2)
    (battery-after b5 b2 b3)
    (battery-after b6 b2 b4)

    ;  Drone
    (at droneA depot1)  (battery droneA b6)  (empty droneA)

    ;  Accessibility & unlocking
    (accessible s1)
    (accessible s2)
    (accessible s3)
    (accessible s5)
    (accessible s6)
    (accessible depot1)
    (accessible depot2)
    (accessible charging1)
    (accessible charging2)
```

```
(accessible charging3)
(unlocks-by-delivery comms1 s3 s4)

;  Charger & supplies
(has-charger charging1)
(has-charger charging2)
(has-charger charging3)
(supply-at medkit1 depot1)
(supply-at medkit2 depot2)
(supply-at comms1 depot2)
(supply-at water1 depot1)
(needs medkit1 s1)
(needs water1 s2)
(needs comms1 s3)
(needs medkit2 s5)

;  Map: each edge needs 6 facts (both directions)
(connected depot1 s1) (connected s1 depot1)
(flight-cost s1 depot1 b1) (flight-cost depot1 s1 b1)
(= (distance s1 depot1) 1) (= (distance depot1 s1) 1)

(connected depot1 s2) (connected s2 depot1)
(flight-cost s2 depot1 b2) (flight-cost depot1 s2 b2)
(= (distance s2 depot1) 2) (= (distance depot1 s2) 2)

(connected depot1 charging1) (connected charging1 depot1)
(flight-cost charging1 depot1 b1) (flight-cost depot1 charging1 b1)
(= (distance charging1 depot1) 1) (= (distance depot1 charging1) 1)

(connected charging1 charging2) (connected charging2 charging1)
(flight-cost charging2 charging1 b2) (flight-cost charging1 charging2 b2)
(= (distance charging1 charging2) 2) (= (distance charging2 charging1) 2)

(connected charging2 s3) (connected s3 charging2)
(flight-cost s3 charging2 b1) (flight-cost charging2 s3 b1)
(= (distance charging2 s3) 1) (= (distance s3 charging2) 1)

(connected s3 s4) (connected s4 s3)
(flight-cost s4 s3 b1) (flight-cost s3 s4 b1)
(= (distance s3 s4) 1) (= (distance s4 s3) 1)

(connected charging2 charging3) (connected charging3 charging2)
(flight-cost charging3 charging2 b2) (flight-cost charging2 charging3 b2)
(= (distance charging2 charging3) 2) (= (distance charging3 charging2) 2)

(connected charging3 depot2) (connected depot2 charging3)
(flight-cost depot2 charging3 b1) (flight-cost charging3 depot2 b1)
(= (distance charging3 depot2) 1) (= (distance depot2 charging3) 1)

(connected depot2 s5) (connected s5 depot2)
(flight-cost s5 depot2 b1) (flight-cost depot2 s5 b1)
```

```
    (= (distance depot2 s5) 1) (= (distance s5 depot2) 1)

    (connected s5 s6) (connected s6 s5)
    (flight-cost s6 s5 b2) (flight-cost s5 s6 b2)
    (= (distance s5 s6) 2) (= (distance s6 s5) 2)

    ; Plan cost
    (= (total-cost) 0)
  )

  (:goal (and
    (inspected s2)
    (inspected s4)
    (inspected s6)
    (delivered medkit1 s1)
    (delivered medkit2 s5)
    (delivered water1 s2)
    (delivered comms1 s3)
  ))

  (:metric minimize (total-cost))
)
```

A3. Example Plans

*A3.1 large-A-A**

None

```
(pick-up-supply dronea medkit1 depot1)
(pick-up-supply dronec comms1 depot2)
(fly dronec depot2 charging3 b6 b1 b5)
(fly droneb charging2 charging3 b6 b2 b4)
(fly droneb charging3 depot2 b4 b1 b3)
(fly dronec charging3 charging2 b5 b2 b3)
(pick-up-supply droneb medkit2 depot2)
(fly droneb depot2 s5 b3 b1 b2)
(deliver-supply droneb medkit2 s5)
(fly droneb s5 s6 b2 b2 b0)
(inspect droneb s6)
(fly dronec charging2 s3 b3 b1 b2)
(deliver-supply dronec comms1 s3)
(fly dronec s3 s4 b2 b1 b1)
(inspect dronec s4)
(fly dronea depot1 s1 b6 b1 b5)
(deliver-supply dronea medkit1 s1)
(fly dronea s1 depot1 b5 b1 b4)
(pick-up-supply dronea water1 depot1)
(fly dronea depot1 s2 b4 b2 b2)
(deliver-supply dronea water1 s2)
```

```
(inspect dronea s2)
; cost = 26 (general cost)
```

A3.2 large-A-lama

```
None
(pick-up-supply dronea medkit1 depot1)
(pick-up-supply dronec medkit2 depot2)
(fly dronea depot1 s1 b6 b1 b5)
(deliver-supply dronea medkit1 s1)
(fly dronea s1 depot1 b5 b1 b4)
(pick-up-supply dronea water1 depot1)
(fly dronea depot1 s2 b4 b2 b2)
(deliver-supply dronea water1 s2)
(inspect dronea s2)
(fly droneb charging2 s3 b6 b1 b5)
(fly dronec depot2 s5 b6 b1 b5)
(deliver-supply dronec medkit2 s5)
(fly dronec s5 s6 b5 b2 b3)
(inspect dronec s6)
(fly droneb s3 charging2 b5 b1 b4)
(fly droneb charging2 charging3 b4 b2 b2)
(fly droneb charging3 charging2 b2 b2 b0)
(recharge droneb charging2 b0 b6)
(fly droneb charging2 charging3 b6 b2 b4)
(fly droneb charging3 depot2 b4 b1 b3)
(pick-up-supply droneb comms1 depot2)
(fly droneb depot2 charging3 b3 b1 b2)
(fly droneb charging3 charging2 b2 b2 b0)
(recharge droneb charging2 b0 b6)
(fly droneb charging2 s3 b6 b1 b5)
(deliver-supply droneb comms1 s3)
(fly droneb s3 s4 b5 b1 b4)
(inspect droneb s4)
; cost = 34 (general cost)
```

*A3.3 large-B-A**

```
None
(pick-up-supply dronea medkit1 depot1)
(pick-up-supply dronec comms1 depot2)
(fly dronec depot2 charging3 b6 b1 b5)
(fly droneb charging2 charging3 b6 b2 b4)
(fly droneb charging3 depot2 b4 b1 b3)
```

```
(fly dronec charging3 charging2 b5 b2 b3)
(pick-up-supply droneb medkit2 depot2)
(fly droneb depot2 s5 b3 b1 b2)
(deliver-supply droneb medkit2 s5)
(fly droneb s5 s6 b2 b2 b0)
(inspect droneb s6)
(fly dronec charging2 s3 b3 b1 b2)
(deliver-supply dronec comms1 s3)
(fly dronec s3 s4 b2 b1 b1)
(inspect dronec s4)
(fly dronea depot1 s1 b6 b1 b5)
(deliver-supply dronea medkit1 s1)
(fly dronea s1 depot1 b5 b1 b4)
(pick-up-supply dronea water1 depot1)
(fly dronea depot1 s2 b4 b2 b2)
(deliver-supply dronea water1 s2)
(inspect dronea s2)
; cost = 26 (general cost)
```

A3.4 large-B-lama

None

```
(pick-up-supply dronea medkit1 depot1)
(pick-up-supply dronec medkit2 depot2)
(fly dronea depot1 s1 b6 b1 b5)
(deliver-supply dronea medkit1 s1)
(fly dronea s1 depot1 b5 b1 b4)
(pick-up-supply dronea water1 depot1)
(fly dronea depot1 s2 b4 b2 b2)
(deliver-supply dronea water1 s2)
(inspect dronea s2)
(fly droneb charging2 s3 b6 b1 b5)
(fly dronec depot2 s5 b6 b1 b5)
(deliver-supply dronec medkit2 s5)
(fly dronec s5 s6 b5 b2 b3)
(inspect dronec s6)
(fly droneb s3 charging2 b5 b1 b4)
(fly droneb charging2 charging3 b4 b2 b2)
(fly droneb charging3 depot2 b2 b1 b1)
(pick-up-supply droneb comms1 depot2)
(fly dronea s2 depot1 b2 b2 b0)
(fly dronec s6 s5 b3 b2 b1)
(fly droneb depot2 charging3 b1 b1 b0)
(recharge droneb charging3 b0 b1)
(recharge droneb charging3 b1 b2)
(fly droneb charging3 charging2 b2 b2 b0)
(recharge droneb charging2 b0 b1)
(recharge droneb charging2 b1 b2)
(fly droneb charging2 s3 b2 b1 b1)
(deliver-supply droneb comms1 s3)
```

```
(fly droneb s3 s4 b1 b1 b0)
(inspect droneb s4)
; cost = 36 (general cost)
```

*A3.5 large1-A-A**

None

```
(pick-up-supply dronea water1 depot1)
(fly dronea depot1 s2 b6 b2 b4)
(deliver-supply dronea water1 s2)
(inspect dronea s2)
(fly dronea s2 depot1 b4 b2 b2)
(pick-up-supply dronea medkit1 depot1)
(fly dronea depot1 charging1 b2 b1 b1)
(recharge dronea charging1 b1 b6)
(fly dronea charging1 depot1 b6 b1 b5)
(fly dronea depot1 s1 b5 b1 b4)
(deliver-supply dronea medkit1 s1)
(fly dronea s1 depot1 b4 b1 b3)
(fly dronea depot1 charging1 b3 b1 b2)
(fly dronea charging1 charging2 b2 b2 b0)
(recharge dronea charging2 b0 b6)
(fly dronea charging2 charging3 b6 b2 b4)
(fly dronea charging3 depot2 b4 b1 b3)
(pick-up-supply dronea comms1 depot2)
(fly dronea depot2 charging3 b3 b1 b2)
(fly dronea charging3 charging2 b2 b2 b0)
(recharge dronea charging2 b0 b6)
(fly dronea charging2 s3 b6 b1 b5)
(deliver-supply dronea comms1 s3)
(fly dronea s3 s4 b5 b1 b4)
(inspect dronea s4)
(fly dronea s4 s3 b4 b1 b3)
(fly dronea s3 charging2 b3 b1 b2)
(fly dronea charging2 charging3 b2 b2 b0)
(recharge dronea charging3 b0 b6)
(fly dronea charging3 depot2 b6 b1 b5)
(pick-up-supply dronea medkit2 depot2)
(fly dronea depot2 s5 b5 b1 b4)
(deliver-supply dronea medkit2 s5)
(fly dronea s5 s6 b4 b2 b2)
(inspect dronea s6)
; cost = 42 (general cost)
```

A3.6 large1-A-lama

None

```
(pick-up-supply dronea medkit1 depot1)
(fly dronea depot1 s1 b6 b1 b5)
```

```
(deliver-supply dronea medkit1 s1)
(fly dronea s1 depot1 b5 b1 b4)
(fly dronea depot1 charging1 b4 b1 b3)
(fly dronea charging1 charging2 b3 b2 b1)
(recharge dronea charging2 b1 b6)
(fly dronea charging2 charging3 b6 b2 b4)
(fly dronea charging3 depot2 b4 b1 b3)
(pick-up-supply dronea medkit2 depot2)
(fly dronea depot2 s5 b3 b1 b2)
(deliver-supply dronea medkit2 s5)
(fly dronea s5 depot2 b2 b1 b1)
(pick-up-supply dronea comms1 depot2)
(fly dronea depot2 charging3 b1 b1 b0)
(recharge dronea charging3 b0 b6)
(fly dronea charging3 charging2 b6 b2 b4)
(fly dronea charging2 s3 b4 b1 b3)
(deliver-supply dronea comms1 s3)
(fly dronea s3 s4 b3 b1 b2)
(inspect dronea s4)
(fly dronea s4 s3 b2 b1 b1)
(fly dronea s3 charging2 b1 b1 b0)
(recharge dronea charging2 b0 b6)
(fly dronea charging2 charging1 b6 b2 b4)
(fly dronea charging1 depot1 b4 b1 b3)
(pick-up-supply dronea water1 depot1)
(fly dronea depot1 charging1 b3 b1 b2)
(fly dronea charging1 depot1 b2 b1 b1)
(fly dronea depot1 charging1 b1 b1 b0)
(recharge dronea charging1 b0 b6)
(fly dronea charging1 depot1 b6 b1 b5)
(fly dronea depot1 s2 b5 b2 b3)
(deliver-supply dronea water1 s2)
(inspect dronea s2)
(fly dronea s2 depot1 b3 b2 b1)
(fly dronea depot1 charging1 b1 b1 b0)
(recharge dronea charging1 b0 b6)
(fly dronea charging1 charging2 b6 b2 b4)
(fly dronea charging2 charging3 b4 b2 b2)
(fly dronea charging3 depot2 b2 b1 b1)
(fly dronea depot2 charging3 b1 b1 b0)
(recharge dronea charging3 b0 b6)
(fly dronea charging3 depot2 b6 b1 b5)
(fly dronea depot2 s5 b5 b1 b4)
(fly dronea s5 s6 b4 b2 b2)
(inspect dronea s6)
; cost = 56 (general cost)
```

*A3.7 large1-B-A**

None

```
(pick-up-supply dronea water1 depot1)
(fly dronea depot1 s2 b6 b2 b4)
(deliver-supply dronea water1 s2)
(inspect dronea s2)
(fly dronea s2 depot1 b4 b2 b2)
(pick-up-supply dronea medkit1 depot1)
(fly dronea depot1 charging1 b2 b1 b1)
(recharge dronea charging1 b1 b2)
(recharge dronea charging1 b2 b3)
(recharge dronea charging1 b3 b4)
(recharge dronea charging1 b4 b5)
(fly dronea charging1 depot1 b5 b1 b4)
(fly dronea depot1 s1 b4 b1 b3)
(deliver-supply dronea medkit1 s1)
(fly dronea s1 depot1 b3 b1 b2)
(fly dronea depot1 charging1 b2 b1 b1)
(recharge dronea charging1 b1 b2)
(recharge dronea charging1 b2 b3)
(fly dronea charging1 charging2 b3 b2 b1)
(recharge dronea charging2 b1 b2)
(recharge dronea charging2 b2 b3)
(recharge dronea charging2 b3 b4)
(recharge dronea charging2 b4 b5)
(recharge dronea charging2 b5 b6)
(fly dronea charging2 charging3 b6 b2 b4)
(fly dronea charging3 depot2 b4 b1 b3)
(pick-up-supply dronea comms1 depot2)
(fly dronea depot2 charging3 b3 b1 b2)
(recharge dronea charging3 b2 b3)
(recharge dronea charging3 b3 b4)
(recharge dronea charging3 b4 b5)
(recharge dronea charging3 b5 b6)
(fly dronea charging3 charging2 b6 b2 b4)
(recharge dronea charging2 b4 b5)
(fly dronea charging2 s3 b5 b1 b4)
(deliver-supply dronea comms1 s3)
(fly dronea s3 s4 b4 b1 b3)
(inspect dronea s4)
(fly dronea s4 s3 b3 b1 b2)
(fly dronea s3 charging2 b2 b1 b1)
(recharge dronea charging2 b1 b2)
(recharge dronea charging2 b2 b3)
(fly dronea charging2 charging3 b3 b2 b1)
(recharge dronea charging3 b1 b2)
(recharge dronea charging3 b2 b3)
(recharge dronea charging3 b3 b4)
(fly dronea charging3 depot2 b4 b1 b3)
(pick-up-supply dronea medkit2 depot2)
(fly dronea depot2 s5 b3 b1 b2)
(deliver-supply dronea medkit2 s5)
```

```
(fly dronea s5 s6 b2 b2 b0)
(inspect dronea s6)
; cost = 59 (general cost)
```

A3.8 large1-B-lama

None

```
(pick-up-supply dronea water1 depot1)
(fly dronea depot1 s2 b6 b2 b4)
(deliver-supply dronea water1 s2)
(inspect dronea s2)
(fly dronea s2 depot1 b4 b2 b2)
(fly dronea depot1 charging1 b2 b1 b1)
(recharge dronea charging1 b1 b2)
(fly dronea charging1 charging2 b2 b2 b0)
(recharge dronea charging2 b0 b1)
(recharge dronea charging2 b1 b2)
(fly dronea charging2 charging3 b2 b2 b0)
(recharge dronea charging3 b0 b1)
(recharge dronea charging3 b1 b2)
(fly dronea charging3 depot2 b2 b1 b1)
(pick-up-supply dronea comms1 depot2)
(fly dronea depot2 charging3 b1 b1 b0)
(recharge dronea charging3 b0 b1)
(recharge dronea charging3 b1 b2)
(fly dronea charging3 charging2 b2 b2 b0)
(recharge dronea charging2 b0 b1)
(recharge dronea charging2 b1 b2)
(fly dronea charging2 s3 b2 b1 b1)
(deliver-supply dronea comms1 s3)
(fly dronea s3 charging2 b1 b1 b0)
(recharge dronea charging2 b0 b1)
(recharge dronea charging2 b1 b2)
(fly dronea charging2 charging1 b2 b2 b0)
(recharge dronea charging1 b0 b1)
(recharge dronea charging1 b1 b2)
(fly dronea charging1 depot1 b2 b1 b1)
(pick-up-supply dronea medkit1 depot1)
(fly dronea depot1 charging1 b1 b1 b0)
(recharge dronea charging1 b0 b1)
(recharge dronea charging1 b1 b2)
(recharge dronea charging1 b2 b3)
(recharge dronea charging1 b3 b4)
(fly dronea charging1 depot1 b4 b1 b3)
(fly dronea depot1 s1 b3 b1 b2)
(deliver-supply dronea medkit1 s1)
(fly dronea s1 depot1 b2 b1 b1)
(fly dronea depot1 charging1 b1 b1 b0)
(recharge dronea charging1 b0 b1)
(recharge dronea charging1 b1 b2)
```

```
(fly dronea charging1 charging2 b2 b2 b0)
(recharge dronea charging2 b0 b1)
(recharge dronea charging2 b1 b2)
(fly dronea charging2 charging3 b2 b2 b0)
(recharge dronea charging3 b0 b1)
(recharge dronea charging3 b1 b2)
(fly dronea charging3 depot2 b2 b1 b1)
(pick-up-supply dronea medkit2 depot2)
(fly dronea depot2 charging3 b1 b1 b0)
(recharge dronea charging3 b0 b1)
(recharge dronea charging3 b1 b2)
(recharge dronea charging3 b2 b3)
(recharge dronea charging3 b3 b4)
(fly dronea charging3 depot2 b4 b1 b3)
(fly dronea depot2 s5 b3 b1 b2)
(deliver-supply dronea medkit2 s5)
(fly dronea s5 depot2 b2 b1 b1)
(fly dronea depot2 charging3 b1 b1 b0)
(recharge dronea charging3 b0 b1)
(recharge dronea charging3 b1 b2)
(fly dronea charging3 charging2 b2 b2 b0)
(recharge dronea charging2 b0 b1)
(recharge dronea charging2 b1 b2)
(recharge dronea charging2 b2 b3)
(recharge dronea charging2 b3 b4)
(fly dronea charging2 s3 b4 b1 b3)
(fly dronea s3 s4 b3 b1 b2)
(inspect dronea s4)
(fly dronea s4 s3 b2 b1 b1)
(fly dronea s3 charging2 b1 b1 b0)
(recharge dronea charging2 b0 b1)
(recharge dronea charging2 b1 b2)
(fly dronea charging2 charging3 b2 b2 b0)
(recharge dronea charging3 b0 b1)
(recharge dronea charging3 b1 b2)
(recharge dronea charging3 b2 b3)
(recharge dronea charging3 b3 b4)
(fly dronea charging3 depot2 b4 b1 b3)
(fly dronea depot2 s5 b3 b1 b2)
(fly dronea s5 s6 b2 b2 b0)
(inspect dronea s6)
; cost = 95 (general cost)
```

*A3.9 medium-A-A**

None

```
(pick-up-supply dronea medkit1 depot)
(fly droneb charging2 s3 b6 b1 b5)
(inspect droneb s3)
(fly droneb s3 depot b5 b2 b3)
```

```
(pick-up-supply droneb water1 depot)
(fly droneb depot s3 b3 b2 b1)
(fly droneb s3 charging2 b1 b1 b0)
(recharge droneb charging2 b0 b6)
(fly droneb charging2 s4 b6 b2 b4)
(inspect droneb s4)
(fly droneb s4 s5 b4 b1 b3)
(inspect droneb s5)
(deliver-supply droneb water1 s5)
(fly dronea depot s1 b6 b1 b5)
(deliver-supply dronea medkit1 s1)
(fly dronea s1 depot b5 b1 b4)
(pick-up-supply dronea medkit2 depot)
(fly dronea depot s1 b4 b1 b3)
(fly dronea s1 s2 b3 b2 b1)
(inspect dronea s2)
(deliver-supply dronea medkit2 s2)
; cost = 25 (general cost)
```

A3.10 medium-A-lama

None

```
(fly dronea depot s1 b6 b1 b5)
(fly droneb charging2 s3 b6 b1 b5)
(inspect droneb s3)
(fly dronea s1 depot b5 b1 b4)
(pick-up-supply dronea medkit1 depot)
(fly dronea depot s1 b4 b1 b3)
(deliver-supply dronea medkit1 s1)
(fly dronea s1 s2 b3 b2 b1)
(inspect dronea s2)
(fly droneb s3 charging2 b5 b1 b4)
(fly droneb charging2 s4 b4 b2 b2)
(inspect droneb s4)
(fly droneb s4 charging2 b2 b2 b0)
(recharge droneb charging2 b0 b6)
(fly droneb charging2 s4 b6 b2 b4)
(fly droneb s4 s5 b4 b1 b3)
(inspect droneb s5)
(fly droneb s5 s4 b3 b1 b2)
(fly droneb s4 charging2 b2 b2 b0)
(recharge droneb charging2 b0 b6)
(fly droneb charging2 s3 b6 b1 b5)
(fly droneb s3 depot b5 b2 b3)
(pick-up-supply droneb water1 depot)
(fly droneb depot s3 b3 b2 b1)
(fly droneb s3 charging2 b1 b1 b0)
(recharge droneb charging2 b0 b6)
(fly droneb charging2 s4 b6 b2 b4)
(fly droneb s4 s5 b4 b1 b3)
```

```
(deliver-supply droneb water1 s5)
(fly droneb s5 s4 b3 b1 b2)
(fly droneb s4 charging2 b2 b2 b0)
(recharge droneb charging2 b0 b6)
(fly droneb charging2 s3 b6 b1 b5)
(fly droneb s3 depot b5 b2 b3)
(pick-up-supply droneb medkit2 depot)
(fly droneb depot s1 b3 b1 b2)
(fly droneb s1 s2 b2 b2 b0)
(deliver-supply droneb medkit2 s2)
; cost = 49 (general cost)
```

*A3.11 medium-B-A**

None

```
(pick-up-supply dronea medkit1 depot)
(fly droneb charging2 s3 b6 b1 b5)
(inspect droneb s3)
(fly droneb s3 depot b5 b2 b3)
(pick-up-supply droneb medkit2 depot)
(fly droneb depot s1 b3 b1 b2)
(fly droneb s1 s2 b2 b2 b0)
(inspect droneb s2)
(deliver-supply droneb medkit2 s2)
(fly dronea depot s1 b6 b1 b5)
(deliver-supply dronea medkit1 s1)
(fly dronea s1 depot b5 b1 b4)
(pick-up-supply dronea water1 depot)
(fly dronea depot s3 b4 b2 b2)
(fly dronea s3 charging2 b2 b1 b1)
(recharge dronea charging2 b1 b2)
(recharge dronea charging2 b2 b3)
(fly dronea charging2 s4 b3 b2 b1)
(inspect dronea s4)
(fly dronea s4 s5 b1 b1 b0)
(inspect dronea s5)
(deliver-supply dronea water1 s5)
; cost = 26 (general cost)
```

A3.12 medium-B-lama

None

```
(pick-up-supply dronea medkit1 depot)
(fly dronea depot s1 b6 b1 b5)
(deliver-supply dronea medkit1 s1)
(fly droneb charging2 s3 b6 b1 b5)
```

```
(inspect droneb s3)
(fly dronea s1 depot b5 b1 b4)
(pick-up-supply dronea medkit2 depot)
(fly dronea depot s1 b4 b1 b3)
(fly dronea s1 s2 b3 b2 b1)
(inspect dronea s2)
(deliver-supply dronea medkit2 s2)
(fly droneb s3 depot b5 b2 b3)
(pick-up-supply droneb water1 depot)
(fly droneb depot s3 b3 b2 b1)
(fly droneb s3 charging2 b1 b1 b0)
(recharge droneb charging2 b0 b1)
(recharge droneb charging2 b1 b2)
(recharge droneb charging2 b2 b3)
(fly droneb charging2 s4 b3 b2 b1)
(inspect droneb s4)
(fly droneb s4 s5 b1 b1 b0)
(inspect droneb s5)
(deliver-supply droneb water1 s5)
; cost = 27 (general cost)
```

*A3.13 small-A-A**

None

```
(pick-up-supply dronea kit1 depot)
(fly dronea depot charging b4 b1 b3)
(fly dronea charging s3 b3 b1 b2)
(inspect dronea s3)
(fly dronea s3 charging b2 b1 b1)
(recharge dronea charging b1 b4)
(fly dronea charging depot b4 b1 b3)
(fly dronea depot s1 b3 b1 b2)
(deliver-supply dronea kit1 s1)
(fly dronea s1 s2 b2 b1 b1)
(inspect dronea s2)
; cost = 11 (unit cost)
```

A3.14 small-A-lama

None

```
(pick-up-supply dronea kit1 depot)
(fly dronea depot charging b4 b1 b3)
(fly dronea charging s3 b3 b1 b2)
(inspect dronea s3)
(fly dronea s3 charging b2 b1 b1)
(recharge dronea charging b1 b4)
(fly dronea charging depot b4 b1 b3)
```

```
(fly dronea depot s1 b3 b1 b2)
(deliver-supply dronea kit1 s1)
(fly dronea s1 s2 b2 b1 b1)
(inspect dronea s2)
; cost = 11 (unit cost)
```

*A3.15 small-B-A**

None

```
(pick-up-supply dronea kit1 depot)
(fly dronea depot charging b4 b1 b3)
(fly dronea charging s3 b3 b1 b2)
(inspect dronea s3)
(fly dronea s3 charging b2 b1 b1)
(recharge dronea charging b1 b2)
(recharge dronea charging b2 b3)
(fly dronea charging depot b3 b1 b2)
(fly dronea depot s1 b2 b1 b1)
(deliver-supply dronea kit1 s1)
(fly dronea s1 s2 b1 b1 b0)
(inspect dronea s2)
; cost = 12 (unit cost)
```

A3.16 small-B-lama

None

```
(pick-up-supply dronea kit1 depot)
(fly dronea depot s1 b4 b1 b3)
(deliver-supply dronea kit1 s1)
(fly dronea s1 depot b3 b1 b2)
(fly dronea depot charging b2 b1 b1)
(recharge dronea charging b1 b2)
(fly dronea charging s3 b2 b1 b1)
(inspect dronea s3)
(fly dronea s3 charging b1 b1 b0)
(recharge dronea charging b0 b1)
(recharge dronea charging b1 b2)
(recharge dronea charging b2 b3)
(fly dronea charging depot b3 b1 b2)
(fly dronea depot s1 b2 b1 b1)
(fly dronea s1 s2 b1 b1 b0)
(inspect dronea s2)
; cost = 16 (unit cost)
```

A4. VAL Output

A4.1 large-A-astar

```
None
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/large-A-astar
Plan to validate:

Plan size: 22
1:
(pick-up-supply dronea medkit1 depot1)

2:
(pick-up-supply dronec comms1 depot2)

3:
(fly dronec depot2 charging3 b6 b1 b5)

4:
(fly droneb charging2 charging3 b6 b2 b4)

5:
(fly droneb charging3 depot2 b4 b1 b3)

6:
(fly dronec charging3 charging2 b5 b2 b3)

7:
(pick-up-supply droneb medkit2 depot2)

8:
(fly droneb depot2 s5 b3 b1 b2)

9:
(deliver-supply droneb medkit2 s5)

10:
(fly droneb s5 s6 b2 b2 b0)

11:
(inspect droneb s6)

12:
(fly dronec charging2 s3 b3 b1 b2)
```

```
13:
(deliver-supply dronec comms1 s3)

14:
(fly dronec s3 s4 b2 b1 b1)

15:
(inspect dronec s4)

16:
(fly dronea depot1 s1 b6 b1 b5)

17:
(deliver-supply dronea medkit1 s1)

18:
(fly dronea s1 depot1 b5 b1 b4)

19:
(pick-up-supply dronea water1 depot1)

20:
(fly dronea depot1 s2 b4 b2 b2)

21:
(deliver-supply dronea water1 s2)

22:
(inspect dronea s2)


Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (supply-at medkit1 depot1)
Deleting (empty dronea)
Adding (carrying dronea medkit1)
Updating (total-cost) (0) by 1 increase

Checking next happening (time 2)
Deleting (supply-at comms1 depot2)
Deleting (empty dronec)
Adding (carrying dronec comms1)
Updating (total-cost) (1) by 1 increase

Checking next happening (time 3)
Deleting (at dronec depot2)
Deleting (battery dronec b6)
Adding (at dronec charging3)
Adding (battery dronec b5)
```

```
Updating (total-cost) (2) by 1 increase

Checking next happening (time 4)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
Adding (at droneb charging3)
Adding (battery droneb b4)
Updating (total-cost) (3) by 2 increase

Checking next happening (time 5)
Deleting (at droneb charging3)
Deleting (battery droneb b4)
Adding (at droneb depot2)
Adding (battery droneb b3)
Updating (total-cost) (5) by 1 increase

Checking next happening (time 6)
Deleting (at dronec charging3)
Deleting (battery dronec b5)
Adding (at dronec charging2)
Adding (battery dronec b3)
Updating (total-cost) (6) by 2 increase

Checking next happening (time 7)
Deleting (supply-at medkit2 depot2)
Deleting (empty droneb)
Adding (carrying droneb medkit2)
Updating (total-cost) (8) by 1 increase

Checking next happening (time 8)
Deleting (at droneb depot2)
Deleting (battery droneb b3)
Adding (at droneb s5)
Adding (battery droneb b2)
Updating (total-cost) (9) by 1 increase

Checking next happening (time 9)
Deleting (carrying droneb medkit2)
Deleting (needs medkit2 s5)
Adding (empty droneb)
Adding (delivered medkit2 s5)
Updating (total-cost) (10) by 1 increase

Checking next happening (time 10)
Deleting (at droneb s5)
Deleting (battery droneb b2)
Adding (at droneb s6)
Adding (battery droneb b0)
Updating (total-cost) (11) by 2 increase

Checking next happening (time 11)
```

```
Adding (inspected s6)
Updating (total-cost) (13) by 1 increase

Checking next happening (time 12)
Deleting (at dronec charging2)
Deleting (battery dronec b3)
Adding (at dronec s3)
Adding (battery dronec b2)
Updating (total-cost) (14) by 1 increase

Checking next happening (time 13)
Deleting (carrying dronec comms1)
Deleting (needs comms1 s3)
Adding (empty dronec)
Adding (delivered comms1 s3)
Adding (accessible s4)
Updating (total-cost) (15) by 1 increase

Checking next happening (time 14)
Deleting (at dronec s3)
Deleting (battery dronec b2)
Adding (at dronec s4)
Adding (battery dronec b1)
Updating (total-cost) (16) by 1 increase

Checking next happening (time 15)
Adding (inspected s4)
Updating (total-cost) (17) by 1 increase

Checking next happening (time 16)
Deleting (at dronea depot1)
Deleting (battery dronea b6)
Adding (at dronea s1)
Adding (battery dronea b5)
Updating (total-cost) (18) by 1 increase

Checking next happening (time 17)
Deleting (carrying dronea medkit1)
Deleting (needs medkit1 s1)
Adding (empty dronea)
Adding (delivered medkit1 s1)
Updating (total-cost) (19) by 1 increase

Checking next happening (time 18)
Deleting (at dronea s1)
Deleting (battery dronea b5)
Adding (at dronea depot1)
Adding (battery dronea b4)
Updating (total-cost) (20) by 1 increase

Checking next happening (time 19)
```

```
Deleting (supply-at water1 depot1)
Deleting (empty dronea)
Adding (carrying dronea water1)
Updating (total-cost) (21) by 1 increase

Checking next happening (time 20)
Deleting (at dronea depot1)
Deleting (battery dronea b4)
Adding (at dronea s2)
Adding (battery dronea b2)
Updating (total-cost) (22) by 2 increase

Checking next happening (time 21)
Deleting (carrying dronea water1)
Deleting (needs water1 s2)
Adding (empty dronea)
Adding (delivered water1 s2)
Updating (total-cost) (24) by 1 increase

Checking next happening (time 22)
Adding (inspected s2)
Updating (total-cost) (25) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 26

Successful plans:
Value: 26
 plans/large-A-astar 26
```

A4.2 large-A-lama

```
None
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/large-A-lama
Plan to validate:

Plan size: 28
1:
(pick-up-supply dronea medkit1 depot1)
```

```
2:
(pick-up-supply dronec medkit2 depot2)

3:
(fly dronea depot1 s1 b6 b1 b5)

4:
(deliver-supply dronea medkit1 s1)

5:
(fly dronea s1 depot1 b5 b1 b4)

6:
(pick-up-supply dronea water1 depot1)

7:
(fly dronea depot1 s2 b4 b2 b2)

8:
(deliver-supply dronea water1 s2)

9:
(inspect dronea s2)

10:
(fly droneb charging2 s3 b6 b1 b5)

11:
(fly dronec depot2 s5 b6 b1 b5)

12:
(deliver-supply dronec medkit2 s5)

13:
(fly dronec s5 s6 b5 b2 b3)

14:
(inspect dronec s6)

15:
(fly droneb s3 charging2 b5 b1 b4)

16:
(fly droneb charging2 charging3 b4 b2 b2)

17:
(fly droneb charging3 charging2 b2 b2 b0)

18:
(recharge droneb charging2 b0 b6)
```

```
19:
(fly droneb charging2 charging3 b6 b2 b4)

20:
(fly droneb charging3 depot2 b4 b1 b3)

21:
(pick-up-supply droneb comms1 depot2)

22:
(fly droneb depot2 charging3 b3 b1 b2)

23:
(fly droneb charging3 charging2 b2 b2 b0)

24:
(recharge droneb charging2 b0 b6)

25:
(fly droneb charging2 s3 b6 b1 b5)

26:
(deliver-supply droneb comms1 s3)

27:
(fly droneb s3 s4 b5 b1 b4)

28:
(inspect droneb s4)


Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (supply-at medkit1 depot1)
Deleting (empty dronea)
Adding (carrying dronea medkit1)
Updating (total-cost) (0) by 1 increase

Checking next happening (time 2)
Deleting (supply-at medkit2 depot2)
Deleting (empty dronec)
Adding (carrying dronec medkit2)
Updating (total-cost) (1) by 1 increase

Checking next happening (time 3)
Deleting (at dronea depot1)
Deleting (battery dronea b6)
Adding (at dronea s1)
```

```
Adding (battery dronea b5)
Updating (total-cost) (2) by 1 increase

Checking next happening (time 4)
Deleting (carrying dronea medkit1)
Deleting (needs medkit1 s1)
Adding (empty dronea)
Adding (delivered medkit1 s1)
Updating (total-cost) (3) by 1 increase

Checking next happening (time 5)
Deleting (at dronea s1)
Deleting (battery dronea b5)
Adding (at dronea depot1)
Adding (battery dronea b4)
Updating (total-cost) (4) by 1 increase

Checking next happening (time 6)
Deleting (supply-at water1 depot1)
Deleting (empty dronea)
Adding (carrying dronea water1)
Updating (total-cost) (5) by 1 increase

Checking next happening (time 7)
Deleting (at dronea depot1)
Deleting (battery dronea b4)
Adding (at dronea s2)
Adding (battery dronea b2)
Updating (total-cost) (6) by 2 increase

Checking next happening (time 8)
Deleting (carrying dronea water1)
Deleting (needs water1 s2)
Adding (empty dronea)
Adding (delivered water1 s2)
Updating (total-cost) (8) by 1 increase

Checking next happening (time 9)
Adding (inspected s2)
Updating (total-cost) (9) by 1 increase

Checking next happening (time 10)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
Adding (at droneb s3)
Adding (battery droneb b5)
Updating (total-cost) (10) by 1 increase

Checking next happening (time 11)
Deleting (at dronec depot2)
Deleting (battery dronec b6)
```

```
Adding (at dronec s5)
Adding (battery dronec b5)
Updating (total-cost) (11) by 1 increase

Checking next happening (time 12)
Deleting (carrying dronec medkit2)
Deleting (needs medkit2 s5)
Adding (empty dronec)
Adding (delivered medkit2 s5)
Updating (total-cost) (12) by 1 increase

Checking next happening (time 13)
Deleting (at dronec s5)
Deleting (battery dronec b5)
Adding (at dronec s6)
Adding (battery dronec b3)
Updating (total-cost) (13) by 2 increase

Checking next happening (time 14)
Adding (inspected s6)
Updating (total-cost) (15) by 1 increase

Checking next happening (time 15)
Deleting (at droneb s3)
Deleting (battery droneb b5)
Adding (at droneb charging2)
Adding (battery droneb b4)
Updating (total-cost) (16) by 1 increase

Checking next happening (time 16)
Deleting (at droneb charging2)
Deleting (battery droneb b4)
Adding (at droneb charging3)
Adding (battery droneb b2)
Updating (total-cost) (17) by 2 increase

Checking next happening (time 17)
Deleting (at droneb charging3)
Deleting (battery droneb b2)
Adding (at droneb charging2)
Adding (battery droneb b0)
Updating (total-cost) (19) by 2 increase

Checking next happening (time 18)
Deleting (battery droneb b0)
Adding (battery droneb b6)
Updating (total-cost) (21) by 1 increase

Checking next happening (time 19)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
```

```
Adding (at droneb charging3)
Adding (battery droneb b4)
Updating (total-cost) (22) by 2 increase

Checking next happening (time 20)
Deleting (at droneb charging3)
Deleting (battery droneb b4)
Adding (at droneb depot2)
Adding (battery droneb b3)
Updating (total-cost) (24) by 1 increase

Checking next happening (time 21)
Deleting (supply-at comms1 depot2)
Deleting (empty droneb)
Adding (carrying droneb comms1)
Updating (total-cost) (25) by 1 increase

Checking next happening (time 22)
Deleting (at droneb depot2)
Deleting (battery droneb b3)
Adding (at droneb charging3)
Adding (battery droneb b2)
Updating (total-cost) (26) by 1 increase

Checking next happening (time 23)
Deleting (at droneb charging3)
Deleting (battery droneb b2)
Adding (at droneb charging2)
Adding (battery droneb b0)
Updating (total-cost) (27) by 2 increase

Checking next happening (time 24)
Deleting (battery droneb b0)
Adding (battery droneb b6)
Updating (total-cost) (29) by 1 increase

Checking next happening (time 25)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
Adding (at droneb s3)
Adding (battery droneb b5)
Updating (total-cost) (30) by 1 increase

Checking next happening (time 26)
Deleting (carrying droneb comms1)
Deleting (needs comms1 s3)
Adding (empty droneb)
Adding (delivered comms1 s3)
Adding (accessible s4)
Updating (total-cost) (31) by 1 increase
```

```
Checking next happening (time 27)
Deleting (at droneb s3)
Deleting (battery droneb b5)
Adding (at droneb s4)
Adding (battery droneb b4)
Updating (total-cost) (32) by 1 increase

Checking next happening (time 28)
Adding (inspected s4)
Updating (total-cost) (33) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 34

Successful plans:
Value: 34
 plans/large-A-lama 34
```

A4.3 large-B-astar

None

```
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/large-B-astar
Plan to validate:

Plan size: 22
1:
(pick-up-supply dronea medkit1 depot1)

2:
(pick-up-supply dronec comms1 depot2)

3:
(fly dronec depot2 charging3 b6 b1 b5)

4:
(fly droneb charging2 charging3 b6 b2 b4)

5:
(fly droneb charging3 depot2 b4 b1 b3)
```

```
6:
(fly dronec charging3 charging2 b5 b2 b3)

7:
(pick-up-supply droneb medkit2 depot2)

8:
(fly droneb depot2 s5 b3 b1 b2)

9:
(deliver-supply droneb medkit2 s5)

10:
(fly droneb s5 s6 b2 b2 b0)

11:
(inspect droneb s6)

12:
(fly dronec charging2 s3 b3 b1 b2)

13:
(deliver-supply dronec comms1 s3)

14:
(fly dronec s3 s4 b2 b1 b1)

15:
(inspect dronec s4)

16:
(fly dronea depot1 s1 b6 b1 b5)

17:
(deliver-supply dronea medkit1 s1)

18:
(fly dronea s1 depot1 b5 b1 b4)

19:
(pick-up-supply dronea water1 depot1)

20:
(fly dronea depot1 s2 b4 b2 b2)

21:
(deliver-supply dronea water1 s2)

22:
(inspect dronea s2)
```

```
Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (supply-at medkit1 depot1)
Deleting (empty dronea)
Adding (carrying dronea medkit1)
Updating (total-cost) (0) by 1 increase

Checking next happening (time 2)
Deleting (supply-at comms1 depot2)
Deleting (empty dronec)
Adding (carrying dronec comms1)
Updating (total-cost) (1) by 1 increase

Checking next happening (time 3)
Deleting (at dronec depot2)
Deleting (battery dronec b6)
Adding (at dronec charging3)
Adding (battery dronec b5)
Updating (total-cost) (2) by 1 increase

Checking next happening (time 4)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
Adding (at droneb charging3)
Adding (battery droneb b4)
Updating (total-cost) (3) by 2 increase

Checking next happening (time 5)
Deleting (at droneb charging3)
Deleting (battery droneb b4)
Adding (at droneb depot2)
Adding (battery droneb b3)
Updating (total-cost) (5) by 1 increase

Checking next happening (time 6)
Deleting (at dronec charging3)
Deleting (battery dronec b5)
Adding (at dronec charging2)
Adding (battery dronec b3)
Updating (total-cost) (6) by 2 increase

Checking next happening (time 7)
Deleting (supply-at medkit2 depot2)
Deleting (empty droneb)
Adding (carrying droneb medkit2)
Updating (total-cost) (8) by 1 increase
```

```
Checking next happening (time 8)
Deleting (at droneb depot2)
Deleting (battery droneb b3)
Adding (at droneb s5)
Adding (battery droneb b2)
Updating (total-cost) (9) by 1 increase

Checking next happening (time 9)
Deleting (carrying droneb medkit2)
Deleting (needs medkit2 s5)
Adding (empty droneb)
Adding (delivered medkit2 s5)
Updating (total-cost) (10) by 1 increase

Checking next happening (time 10)
Deleting (at droneb s5)
Deleting (battery droneb b2)
Adding (at droneb s6)
Adding (battery droneb b0)
Updating (total-cost) (11) by 2 increase

Checking next happening (time 11)
Adding (inspected s6)
Updating (total-cost) (13) by 1 increase

Checking next happening (time 12)
Deleting (at dronec charging2)
Deleting (battery dronec b3)
Adding (at dronec s3)
Adding (battery dronec b2)
Updating (total-cost) (14) by 1 increase

Checking next happening (time 13)
Deleting (carrying dronec comms1)
Deleting (needs comms1 s3)
Adding (empty dronec)
Adding (delivered comms1 s3)
Adding (accessible s4)
Updating (total-cost) (15) by 1 increase

Checking next happening (time 14)
Deleting (at dronec s3)
Deleting (battery dronec b2)
Adding (at dronec s4)
Adding (battery dronec b1)
Updating (total-cost) (16) by 1 increase

Checking next happening (time 15)
Adding (inspected s4)
Updating (total-cost) (17) by 1 increase
```

```
Checking next happening (time 16)
Deleting (at dronea depot1)
Deleting (battery dronea b6)
Adding (at dronea s1)
Adding (battery dronea b5)
Updating (total-cost) (18) by 1 increase

Checking next happening (time 17)
Deleting (carrying dronea medkit1)
Deleting (needs medkit1 s1)
Adding (empty dronea)
Adding (delivered medkit1 s1)
Updating (total-cost) (19) by 1 increase

Checking next happening (time 18)
Deleting (at dronea s1)
Deleting (battery dronea b5)
Adding (at dronea depot1)
Adding (battery dronea b4)
Updating (total-cost) (20) by 1 increase

Checking next happening (time 19)
Deleting (supply-at water1 depot1)
Deleting (empty dronea)
Adding (carrying dronea water1)
Updating (total-cost) (21) by 1 increase

Checking next happening (time 20)
Deleting (at dronea depot1)
Deleting (battery dronea b4)
Adding (at dronea s2)
Adding (battery dronea b2)
Updating (total-cost) (22) by 2 increase

Checking next happening (time 21)
Deleting (carrying dronea water1)
Deleting (needs water1 s2)
Adding (empty dronea)
Adding (delivered water1 s2)
Updating (total-cost) (24) by 1 increase

Checking next happening (time 22)
Adding (inspected s2)
Updating (total-cost) (25) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 26

Successful plans:
Value: 26
 plans/large-B-astar 26
```

A4.4 large-B-lama

```
None
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/large-B-lama
Plan to validate:

Plan size: 30
1:
(pick-up-supply dronea medkit1 depot1)

2:
(pick-up-supply dronec medkit2 depot2)

3:
(fly dronea depot1 s1 b6 b1 b5)

4:
(deliver-supply dronea medkit1 s1)

5:
(fly dronea s1 depot1 b5 b1 b4)

6:
(pick-up-supply dronea water1 depot1)

7:
(fly dronea depot1 s2 b4 b2 b2)

8:
(deliver-supply dronea water1 s2)

9:
(inspect dronea s2)

10:
(fly droneb charging2 s3 b6 b1 b5)

11:
```

```
(fly dronec depot2 s5 b6 b1 b5)

12:
(deliver-supply dronec medkit2 s5)

13:
(fly dronec s5 s6 b5 b2 b3)

14:
(inspect dronec s6)

15:
(fly droneb s3 charging2 b5 b1 b4)

16:
(fly droneb charging2 charging3 b4 b2 b2)

17:
(fly droneb charging3 depot2 b2 b1 b1)

18:
(pick-up-supply droneb comms1 depot2)

19:
(fly dronea s2 depot1 b2 b2 b0)

20:
(fly dronec s6 s5 b3 b2 b1)

21:
(fly droneb depot2 charging3 b1 b1 b0)

22:
(recharge droneb charging3 b0 b1)

23:
(recharge droneb charging3 b1 b2)

24:
(fly droneb charging3 charging2 b2 b2 b0)

25:
(recharge droneb charging2 b0 b1)

26:
(recharge droneb charging2 b1 b2)

27:
(fly droneb charging2 s3 b2 b1 b1)

28:
```

```
(deliver-supply droneb comms1 s3)

29:
(fly droneb s3 s4 b1 b1 b0)

30:
(inspect droneb s4)


Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (supply-at medkit1 depot1)
Deleting (empty dronea)
Adding (carrying dronea medkit1)
Updating (total-cost) (0) by 1 increase

Checking next happening (time 2)
Deleting (supply-at medkit2 depot2)
Deleting (empty dronec)
Adding (carrying dronec medkit2)
Updating (total-cost) (1) by 1 increase

Checking next happening (time 3)
Deleting (at dronea depot1)
Deleting (battery dronea b6)
Adding (at dronea s1)
Adding (battery dronea b5)
Updating (total-cost) (2) by 1 increase

Checking next happening (time 4)
Deleting (carrying dronea medkit1)
Deleting (needs medkit1 s1)
Adding (empty dronea)
Adding (delivered medkit1 s1)
Updating (total-cost) (3) by 1 increase

Checking next happening (time 5)
Deleting (at dronea s1)
Deleting (battery dronea b5)
Adding (at dronea depot1)
Adding (battery dronea b4)
Updating (total-cost) (4) by 1 increase

Checking next happening (time 6)
Deleting (supply-at water1 depot1)
Deleting (empty dronea)
Adding (carrying dronea water1)
Updating (total-cost) (5) by 1 increase
```

```
Checking next happening (time 7)
Deleting (at dronea depot1)
Deleting (battery dronea b4)
Adding (at dronea s2)
Adding (battery dronea b2)
Updating (total-cost) (6) by 2 increase

Checking next happening (time 8)
Deleting (carrying dronea water1)
Deleting (needs water1 s2)
Adding (empty dronea)
Adding (delivered water1 s2)
Updating (total-cost) (8) by 1 increase

Checking next happening (time 9)
Adding (inspected s2)
Updating (total-cost) (9) by 1 increase

Checking next happening (time 10)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
Adding (at droneb s3)
Adding (battery droneb b5)
Updating (total-cost) (10) by 1 increase

Checking next happening (time 11)
Deleting (at dronec depot2)
Deleting (battery dronec b6)
Adding (at dronec s5)
Adding (battery dronec b5)
Updating (total-cost) (11) by 1 increase

Checking next happening (time 12)
Deleting (carrying dronec medkit2)
Deleting (needs medkit2 s5)
Adding (empty dronec)
Adding (delivered medkit2 s5)
Updating (total-cost) (12) by 1 increase

Checking next happening (time 13)
Deleting (at dronec s5)
Deleting (battery dronec b5)
Adding (at dronec s6)
Adding (battery dronec b3)
Updating (total-cost) (13) by 2 increase

Checking next happening (time 14)
Adding (inspected s6)
Updating (total-cost) (15) by 1 increase

Checking next happening (time 15)
```

```
Deleting (at droneb s3)
Deleting (battery droneb b5)
Adding (at droneb charging2)
Adding (battery droneb b4)
Updating (total-cost) (16) by 1 increase

Checking next happening (time 16)
Deleting (at droneb charging2)
Deleting (battery droneb b4)
Adding (at droneb charging3)
Adding (battery droneb b2)
Updating (total-cost) (17) by 2 increase

Checking next happening (time 17)
Deleting (at droneb charging3)
Deleting (battery droneb b2)
Adding (at droneb depot2)
Adding (battery droneb b1)
Updating (total-cost) (19) by 1 increase

Checking next happening (time 18)
Deleting (supply-at comms1 depot2)
Deleting (empty droneb)
Adding (carrying droneb comms1)
Updating (total-cost) (20) by 1 increase

Checking next happening (time 19)
Deleting (at dronea s2)
Deleting (battery dronea b2)
Adding (at dronea depot1)
Adding (battery dronea b0)
Updating (total-cost) (21) by 2 increase

Checking next happening (time 20)
Deleting (at dronec s6)
Deleting (battery dronec b3)
Adding (at dronec s5)
Adding (battery dronec b1)
Updating (total-cost) (23) by 2 increase

Checking next happening (time 21)
Deleting (at droneb depot2)
Deleting (battery droneb b1)
Adding (at droneb charging3)
Adding (battery droneb b0)
Updating (total-cost) (25) by 1 increase

Checking next happening (time 22)
Deleting (battery droneb b0)
Adding (battery droneb b1)
Updating (total-cost) (26) by 1 increase
```

```
Checking next happening (time 23)
Deleting (battery droneb b1)
Adding (battery droneb b2)
Updating (total-cost) (27) by 1 increase

Checking next happening (time 24)
Deleting (at droneb charging3)
Deleting (battery droneb b2)
Adding (at droneb charging2)
Adding (battery droneb b0)
Updating (total-cost) (28) by 2 increase

Checking next happening (time 25)
Deleting (battery droneb b0)
Adding (battery droneb b1)
Updating (total-cost) (30) by 1 increase

Checking next happening (time 26)
Deleting (battery droneb b1)
Adding (battery droneb b2)
Updating (total-cost) (31) by 1 increase

Checking next happening (time 27)
Deleting (at droneb charging2)
Deleting (battery droneb b2)
Adding (at droneb s3)
Adding (battery droneb b1)
Updating (total-cost) (32) by 1 increase

Checking next happening (time 28)
Deleting (carrying droneb comms1)
Deleting (needs comms1 s3)
Adding (empty droneb)
Adding (delivered comms1 s3)
Adding (accessible s4)
Updating (total-cost) (33) by 1 increase

Checking next happening (time 29)
Deleting (at droneb s3)
Deleting (battery droneb b1)
Adding (at droneb s4)
Adding (battery droneb b0)
Updating (total-cost) (34) by 1 increase

Checking next happening (time 30)
Adding (inspected s4)
Updating (total-cost) (35) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 36
```

```
Successful plans:
Value: 36
 plans/large-B-lama 36
```

A4.5 large1-A-astar

```
None
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/large-1drone-A-astar
Plan to validate:

Plan size: 35
1:
(pick-up-supply dronea water1 depot1)

2:
(fly dronea depot1 s2 b6 b2 b4)

3:
(deliver-supply dronea water1 s2)

4:
(inspect dronea s2)

5:
(fly dronea s2 depot1 b4 b2 b2)

6:
(pick-up-supply dronea medkit1 depot1)

7:
(fly dronea depot1 charging1 b2 b1 b1)

8:
(recharge dronea charging1 b1 b6)

9:
(fly dronea charging1 depot1 b6 b1 b5)
```

```
10:
(fly dronea depot1 s1 b5 b1 b4)

11:
(deliver-supply dronea medkit1 s1)

12:
(fly dronea s1 depot1 b4 b1 b3)

13:
(fly dronea depot1 charging1 b3 b1 b2)

14:
(fly dronea charging1 charging2 b2 b2 b0)

15:
(recharge dronea charging2 b0 b6)

16:
(fly dronea charging2 charging3 b6 b2 b4)

17:
(fly dronea charging3 depot2 b4 b1 b3)

18:
(pick-up-supply dronea comms1 depot2)

19:
(fly dronea depot2 charging3 b3 b1 b2)

20:
(fly dronea charging3 charging2 b2 b2 b0)

21:
(recharge dronea charging2 b0 b6)

22:
(fly dronea charging2 s3 b6 b1 b5)

23:
(deliver-supply dronea comms1 s3)

24:
(fly dronea s3 s4 b5 b1 b4)

25:
(inspect dronea s4)

26:
(fly dronea s4 s3 b4 b1 b3)
```

```
27:
(fly dronea s3 charging2 b3 b1 b2)

28:
(fly dronea charging2 charging3 b2 b2 b0)

29:
(recharge dronea charging3 b0 b6)

30:
(fly dronea charging3 depot2 b6 b1 b5)

31:
(pick-up-supply dronea medkit2 depot2)

32:
(fly dronea depot2 s5 b5 b1 b4)

33:
(deliver-supply dronea medkit2 s5)

34:
(fly dronea s5 s6 b4 b2 b2)

35:
(inspect dronea s6)


Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (supply-at water1 depot1)
Deleting (empty dronea)
Adding (carrying dronea water1)
Updating (total-cost) (0) by 1 increase

Checking next happening (time 2)
Deleting (at dronea depot1)
Deleting (battery dronea b6)
Adding (at dronea s2)
Adding (battery dronea b4)
Updating (total-cost) (1) by 2 increase

Checking next happening (time 3)
Deleting (carrying dronea water1)
Deleting (needs water1 s2)
Adding (empty dronea)
Adding (delivered water1 s2)
Updating (total-cost) (3) by 1 increase
```

```
Checking next happening (time 4)
Adding (inspected s2)
Updating (total-cost) (4) by 1 increase

Checking next happening (time 5)
Deleting (at dronea s2)
Deleting (battery dronea b4)
Adding (at dronea depot1)
Adding (battery dronea b2)
Updating (total-cost) (5) by 2 increase

Checking next happening (time 6)
Deleting (supply-at medkit1 depot1)
Deleting (empty dronea)
Adding (carrying dronea medkit1)
Updating (total-cost) (7) by 1 increase

Checking next happening (time 7)
Deleting (at dronea depot1)
Deleting (battery dronea b2)
Adding (at dronea charging1)
Adding (battery dronea b1)
Updating (total-cost) (8) by 1 increase

Checking next happening (time 8)
Deleting (battery dronea b1)
Adding (battery dronea b6)
Updating (total-cost) (9) by 1 increase

Checking next happening (time 9)
Deleting (at dronea charging1)
Deleting (battery dronea b6)
Adding (at dronea depot1)
Adding (battery dronea b5)
Updating (total-cost) (10) by 1 increase

Checking next happening (time 10)
Deleting (at dronea depot1)
Deleting (battery dronea b5)
Adding (at dronea s1)
Adding (battery dronea b4)
Updating (total-cost) (11) by 1 increase

Checking next happening (time 11)
Deleting (carrying dronea medkit1)
Deleting (needs medkit1 s1)
Adding (empty dronea)
Adding (delivered medkit1 s1)
Updating (total-cost) (12) by 1 increase

Checking next happening (time 12)
```

```
Deleting (at dronea s1)
Deleting (battery dronea b4)
Adding (at dronea depot1)
Adding (battery dronea b3)
Updating (total-cost) (13) by 1 increase

Checking next happening (time 13)
Deleting (at dronea depot1)
Deleting (battery dronea b3)
Adding (at dronea charging1)
Adding (battery dronea b2)
Updating (total-cost) (14) by 1 increase

Checking next happening (time 14)
Deleting (at dronea charging1)
Deleting (battery dronea b2)
Adding (at dronea charging2)
Adding (battery dronea b0)
Updating (total-cost) (15) by 2 increase

Checking next happening (time 15)
Deleting (battery dronea b0)
Adding (battery dronea b6)
Updating (total-cost) (17) by 1 increase

Checking next happening (time 16)
Deleting (at dronea charging2)
Deleting (battery dronea b6)
Adding (at dronea charging3)
Adding (battery dronea b4)
Updating (total-cost) (18) by 2 increase

Checking next happening (time 17)
Deleting (at dronea charging3)
Deleting (battery dronea b4)
Adding (at dronea depot2)
Adding (battery dronea b3)
Updating (total-cost) (20) by 1 increase

Checking next happening (time 18)
Deleting (supply-at comms1 depot2)
Deleting (empty dronea)
Adding (carrying dronea comms1)
Updating (total-cost) (21) by 1 increase

Checking next happening (time 19)
Deleting (at dronea depot2)
Deleting (battery dronea b3)
Adding (at dronea charging3)
Adding (battery dronea b2)
Updating (total-cost) (22) by 1 increase
```

```
Checking next happening (time 20)
Deleting (at dronea charging3)
Deleting (battery dronea b2)
Adding (at dronea charging2)
Adding (battery dronea b0)
Updating (total-cost) (23) by 2 increase

Checking next happening (time 21)
Deleting (battery dronea b0)
Adding (battery dronea b6)
Updating (total-cost) (25) by 1 increase

Checking next happening (time 22)
Deleting (at dronea charging2)
Deleting (battery dronea b6)
Adding (at dronea s3)
Adding (battery dronea b5)
Updating (total-cost) (26) by 1 increase

Checking next happening (time 23)
Deleting (carrying dronea comms1)
Deleting (needs comms1 s3)
Adding (empty dronea)
Adding (delivered comms1 s3)
Adding (accessible s4)
Updating (total-cost) (27) by 1 increase

Checking next happening (time 24)
Deleting (at dronea s3)
Deleting (battery dronea b5)
Adding (at dronea s4)
Adding (battery dronea b4)
Updating (total-cost) (28) by 1 increase

Checking next happening (time 25)
Adding (inspected s4)
Updating (total-cost) (29) by 1 increase

Checking next happening (time 26)
Deleting (at dronea s4)
Deleting (battery dronea b4)
Adding (at dronea s3)
Adding (battery dronea b3)
Updating (total-cost) (30) by 1 increase

Checking next happening (time 27)
Deleting (at dronea s3)
Deleting (battery dronea b3)
Adding (at dronea charging2)
Adding (battery dronea b2)
```

```
Updating (total-cost) (31) by 1 increase

Checking next happening (time 28)
Deleting (at dronea charging2)
Deleting (battery dronea b2)
Adding (at dronea charging3)
Adding (battery dronea b0)
Updating (total-cost) (32) by 2 increase

Checking next happening (time 29)
Deleting (battery dronea b0)
Adding (battery dronea b6)
Updating (total-cost) (34) by 1 increase

Checking next happening (time 30)
Deleting (at dronea charging3)
Deleting (battery dronea b6)
Adding (at dronea depot2)
Adding (battery dronea b5)
Updating (total-cost) (35) by 1 increase

Checking next happening (time 31)
Deleting (supply-at medkit2 depot2)
Deleting (empty dronea)
Adding (carrying dronea medkit2)
Updating (total-cost) (36) by 1 increase

Checking next happening (time 32)
Deleting (at dronea depot2)
Deleting (battery dronea b5)
Adding (at dronea s5)
Adding (battery dronea b4)
Updating (total-cost) (37) by 1 increase

Checking next happening (time 33)
Deleting (carrying dronea medkit2)
Deleting (needs medkit2 s5)
Adding (empty dronea)
Adding (delivered medkit2 s5)
Updating (total-cost) (38) by 1 increase

Checking next happening (time 34)
Deleting (at dronea s5)
Deleting (battery dronea b4)
Adding (at dronea s6)
Adding (battery dronea b2)
Updating (total-cost) (39) by 2 increase

Checking next happening (time 35)
Adding (inspected s6)
Updating (total-cost) (41) by 1 increase
```

```
Plan executed successfully - checking goal
Plan valid
Final value: 42

Successful plans:
Value: 42
 plans/large-1drone-A-astar 42
```

A4.6 large1-A-lama

None

```
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/large-1drone-A-lama
Plan to validate:

Plan size: 47
1:
(pick-up-supply dronea medkit1 depot1)

2:
(fly dronea depot1 s1 b6 b1 b5)

3:
(deliver-supply dronea medkit1 s1)

4:
(fly dronea s1 depot1 b5 b1 b4)

5:
(fly dronea depot1 charging1 b4 b1 b3)

6:
(fly dronea charging1 charging2 b3 b2 b1)

7:
(recharge dronea charging2 b1 b6)

8:
(fly dronea charging2 charging3 b6 b2 b4)
```

```
9:
(fly dronea charging3 depot2 b4 b1 b3)

10:
(pick-up-supply dronea medkit2 depot2)

11:
(fly dronea depot2 s5 b3 b1 b2)

12:
(deliver-supply dronea medkit2 s5)

13:
(fly dronea s5 depot2 b2 b1 b1)

14:
(pick-up-supply dronea comms1 depot2)

15:
(fly dronea depot2 charging3 b1 b1 b0)

16:
(recharge dronea charging3 b0 b6)

17:
(fly dronea charging3 charging2 b6 b2 b4)

18:
(fly dronea charging2 s3 b4 b1 b3)

19:
(deliver-supply dronea comms1 s3)

20:
(fly dronea s3 s4 b3 b1 b2)

21:
(inspect dronea s4)

22:
(fly dronea s4 s3 b2 b1 b1)

23:
(fly dronea s3 charging2 b1 b1 b0)

24:
(recharge dronea charging2 b0 b6)

25:
(fly dronea charging2 charging1 b6 b2 b4)
```

```
26:
(fly dronea charging1 depot1 b4 b1 b3)

27:
(pick-up-supply dronea water1 depot1)

28:
(fly dronea depot1 charging1 b3 b1 b2)

29:
(fly dronea charging1 depot1 b2 b1 b1)

30:
(fly dronea depot1 charging1 b1 b1 b0)

31:
(recharge dronea charging1 b0 b6)

32:
(fly dronea charging1 depot1 b6 b1 b5)

33:
(fly dronea depot1 s2 b5 b2 b3)

34:
(deliver-supply dronea water1 s2)

35:
(inspect dronea s2)

36:
(fly dronea s2 depot1 b3 b2 b1)

37:
(fly dronea depot1 charging1 b1 b1 b0)

38:
(recharge dronea charging1 b0 b6)

39:
(fly dronea charging1 charging2 b6 b2 b4)

40:
(fly dronea charging2 charging3 b4 b2 b2)

41:
(fly dronea charging3 depot2 b2 b1 b1)

42:
(fly dronea depot2 charging3 b1 b1 b0)
```

```
43:
(recharge dronea charging3 b0 b6)

44:
(fly dronea charging3 depot2 b6 b1 b5)

45:
(fly dronea depot2 s5 b5 b1 b4)

46:
(fly dronea s5 s6 b4 b2 b2)

47:
(inspect dronea s6)


Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (supply-at medkit1 depot1)
Deleting (empty dronea)
Adding (carrying dronea medkit1)
Updating (total-cost) (0) by 1 increase

Checking next happening (time 2)
Deleting (at dronea depot1)
Deleting (battery dronea b6)
Adding (at dronea s1)
Adding (battery dronea b5)
Updating (total-cost) (1) by 1 increase

Checking next happening (time 3)
Deleting (carrying dronea medkit1)
Deleting (needs medkit1 s1)
Adding (empty dronea)
Adding (delivered medkit1 s1)
Updating (total-cost) (2) by 1 increase

Checking next happening (time 4)
Deleting (at dronea s1)
Deleting (battery dronea b5)
Adding (at dronea depot1)
Adding (battery dronea b4)
Updating (total-cost) (3) by 1 increase

Checking next happening (time 5)
Deleting (at dronea depot1)
Deleting (battery dronea b4)
Adding (at dronea charging1)
Adding (battery dronea b3)
```

```
Updating (total-cost) (4) by 1 increase

Checking next happening (time 6)
Deleting (at dronea charging1)
Deleting (battery dronea b3)
Adding (at dronea charging2)
Adding (battery dronea b1)
Updating (total-cost) (5) by 2 increase

Checking next happening (time 7)
Deleting (battery dronea b1)
Adding (battery dronea b6)
Updating (total-cost) (7) by 1 increase

Checking next happening (time 8)
Deleting (at dronea charging2)
Deleting (battery dronea b6)
Adding (at dronea charging3)
Adding (battery dronea b4)
Updating (total-cost) (8) by 2 increase

Checking next happening (time 9)
Deleting (at dronea charging3)
Deleting (battery dronea b4)
Adding (at dronea depot2)
Adding (battery dronea b3)
Updating (total-cost) (10) by 1 increase

Checking next happening (time 10)
Deleting (supply-at medkit2 depot2)
Deleting (empty dronea)
Adding (carrying dronea medkit2)
Updating (total-cost) (11) by 1 increase

Checking next happening (time 11)
Deleting (at dronea depot2)
Deleting (battery dronea b3)
Adding (at dronea s5)
Adding (battery dronea b2)
Updating (total-cost) (12) by 1 increase

Checking next happening (time 12)
Deleting (carrying dronea medkit2)
Deleting (needs medkit2 s5)
Adding (empty dronea)
Adding (delivered medkit2 s5)
Updating (total-cost) (13) by 1 increase

Checking next happening (time 13)
Deleting (at dronea s5)
Deleting (battery dronea b2)
```

```
Adding (at dronea depot2)
Adding (battery dronea b1)
Updating (total-cost) (14) by 1 increase

Checking next happening (time 14)
Deleting (supply-at comms1 depot2)
Deleting (empty dronea)
Adding (carrying dronea comms1)
Updating (total-cost) (15) by 1 increase

Checking next happening (time 15)
Deleting (at dronea depot2)
Deleting (battery dronea b1)
Adding (at dronea charging3)
Adding (battery dronea b0)
Updating (total-cost) (16) by 1 increase

Checking next happening (time 16)
Deleting (battery dronea b0)
Adding (battery dronea b6)
Updating (total-cost) (17) by 1 increase

Checking next happening (time 17)
Deleting (at dronea charging3)
Deleting (battery dronea b6)
Adding (at dronea charging2)
Adding (battery dronea b4)
Updating (total-cost) (18) by 2 increase

Checking next happening (time 18)
Deleting (at dronea charging2)
Deleting (battery dronea b4)
Adding (at dronea s3)
Adding (battery dronea b3)
Updating (total-cost) (20) by 1 increase

Checking next happening (time 19)
Deleting (carrying dronea comms1)
Deleting (needs comms1 s3)
Adding (empty dronea)
Adding (delivered comms1 s3)
Adding (accessible s4)
Updating (total-cost) (21) by 1 increase

Checking next happening (time 20)
Deleting (at dronea s3)
Deleting (battery dronea b3)
Adding (at dronea s4)
Adding (battery dronea b2)
Updating (total-cost) (22) by 1 increase
```

```
Checking next happening (time 21)
Adding (inspected s4)
Updating (total-cost) (23) by 1 increase

Checking next happening (time 22)
Deleting (at dronea s4)
Deleting (battery dronea b2)
Adding (at dronea s3)
Adding (battery dronea b1)
Updating (total-cost) (24) by 1 increase

Checking next happening (time 23)
Deleting (at dronea s3)
Deleting (battery dronea b1)
Adding (at dronea charging2)
Adding (battery dronea b0)
Updating (total-cost) (25) by 1 increase

Checking next happening (time 24)
Deleting (battery dronea b0)
Adding (battery dronea b6)
Updating (total-cost) (26) by 1 increase

Checking next happening (time 25)
Deleting (at dronea charging2)
Deleting (battery dronea b6)
Adding (at dronea charging1)
Adding (battery dronea b4)
Updating (total-cost) (27) by 2 increase

Checking next happening (time 26)
Deleting (at dronea charging1)
Deleting (battery dronea b4)
Adding (at dronea depot1)
Adding (battery dronea b3)
Updating (total-cost) (29) by 1 increase

Checking next happening (time 27)
Deleting (supply-at water1 depot1)
Deleting (empty dronea)
Adding (carrying dronea water1)
Updating (total-cost) (30) by 1 increase

Checking next happening (time 28)
Deleting (at dronea depot1)
Deleting (battery dronea b3)
Adding (at dronea charging1)
Adding (battery dronea b2)
Updating (total-cost) (31) by 1 increase

Checking next happening (time 29)
```

```
Deleting (at dronea charging1)
Deleting (battery dronea b2)
Adding (at dronea depot1)
Adding (battery dronea b1)
Updating (total-cost) (32) by 1 increase

Checking next happening (time 30)
Deleting (at dronea depot1)
Deleting (battery dronea b1)
Adding (at dronea charging1)
Adding (battery dronea b0)
Updating (total-cost) (33) by 1 increase

Checking next happening (time 31)
Deleting (battery dronea b0)
Adding (battery dronea b6)
Updating (total-cost) (34) by 1 increase

Checking next happening (time 32)
Deleting (at dronea charging1)
Deleting (battery dronea b6)
Adding (at dronea depot1)
Adding (battery dronea b5)
Updating (total-cost) (35) by 1 increase

Checking next happening (time 33)
Deleting (at dronea depot1)
Deleting (battery dronea b5)
Adding (at dronea s2)
Adding (battery dronea b3)
Updating (total-cost) (36) by 2 increase

Checking next happening (time 34)
Deleting (carrying dronea water1)
Deleting (needs water1 s2)
Adding (empty dronea)
Adding (delivered water1 s2)
Updating (total-cost) (38) by 1 increase

Checking next happening (time 35)
Adding (inspected s2)
Updating (total-cost) (39) by 1 increase

Checking next happening (time 36)
Deleting (at dronea s2)
Deleting (battery dronea b3)
Adding (at dronea depot1)
Adding (battery dronea b1)
Updating (total-cost) (40) by 2 increase

Checking next happening (time 37)
```

```
Deleting (at dronea depot1)
Deleting (battery dronea b1)
Adding (at dronea charging1)
Adding (battery dronea b0)
Updating (total-cost) (42) by 1 increase

Checking next happening (time 38)
Deleting (battery dronea b0)
Adding (battery dronea b6)
Updating (total-cost) (43) by 1 increase

Checking next happening (time 39)
Deleting (at dronea charging1)
Deleting (battery dronea b6)
Adding (at dronea charging2)
Adding (battery dronea b4)
Updating (total-cost) (44) by 2 increase

Checking next happening (time 40)
Deleting (at dronea charging2)
Deleting (battery dronea b4)
Adding (at dronea charging3)
Adding (battery dronea b2)
Updating (total-cost) (46) by 2 increase

Checking next happening (time 41)
Deleting (at dronea charging3)
Deleting (battery dronea b2)
Adding (at dronea depot2)
Adding (battery dronea b1)
Updating (total-cost) (48) by 1 increase

Checking next happening (time 42)
Deleting (at dronea depot2)
Deleting (battery dronea b1)
Adding (at dronea charging3)
Adding (battery dronea b0)
Updating (total-cost) (49) by 1 increase

Checking next happening (time 43)
Deleting (battery dronea b0)
Adding (battery dronea b6)
Updating (total-cost) (50) by 1 increase

Checking next happening (time 44)
Deleting (at dronea charging3)
Deleting (battery dronea b6)
Adding (at dronea depot2)
Adding (battery dronea b5)
Updating (total-cost) (51) by 1 increase
```

```
Checking next happening (time 45)
Deleting (at dronea depot2)
Deleting (battery dronea b5)
Adding (at dronea s5)
Adding (battery dronea b4)
Updating (total-cost) (52) by 1 increase

Checking next happening (time 46)
Deleting (at dronea s5)
Deleting (battery dronea b4)
Adding (at dronea s6)
Adding (battery dronea b2)
Updating (total-cost) (53) by 2 increase

Checking next happening (time 47)
Adding (inspected s6)
Updating (total-cost) (55) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 56

Successful plans:
Value: 56
 plans/large-1drone-A-lama 56
```

A4.7 large1-B-astar

None

```
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/large-1drone-B-astar
Plan to validate:

Plan size: 52
1:
(pick-up-supply dronea water1 depot1)

2:
(fly dronea depot1 s2 b6 b2 b4)

3:
```

```
(deliver-supply dronea water1 s2)

4:
(inspect dronea s2)

5:
(fly dronea s2 depot1 b4 b2 b2)

6:
(pick-up-supply dronea medkit1 depot1)

7:
(fly dronea depot1 charging1 b2 b1 b1)

8:
(recharge dronea charging1 b1 b2)

9:
(recharge dronea charging1 b2 b3)

10:
(recharge dronea charging1 b3 b4)

11:
(recharge dronea charging1 b4 b5)

12:
(fly dronea charging1 depot1 b5 b1 b4)

13:
(fly dronea depot1 s1 b4 b1 b3)

14:
(deliver-supply dronea medkit1 s1)

15:
(fly dronea s1 depot1 b3 b1 b2)

16:
(fly dronea depot1 charging1 b2 b1 b1)

17:
(recharge dronea charging1 b1 b2)

18:
(recharge dronea charging1 b2 b3)

19:
(fly dronea charging1 charging2 b3 b2 b1)

20:
```

```
(recharge dronea charging2 b1 b2)

21:
(recharge dronea charging2 b2 b3)

22:
(recharge dronea charging2 b3 b4)

23:
(recharge dronea charging2 b4 b5)

24:
(recharge dronea charging2 b5 b6)

25:
(fly dronea charging2 charging3 b6 b2 b4)

26:
(fly dronea charging3 depot2 b4 b1 b3)

27:
(pick-up-supply dronea comms1 depot2)

28:
(fly dronea depot2 charging3 b3 b1 b2)

29:
(recharge dronea charging3 b2 b3)

30:
(recharge dronea charging3 b3 b4)

31:
(recharge dronea charging3 b4 b5)

32:
(recharge dronea charging3 b5 b6)

33:
(fly dronea charging3 charging2 b6 b2 b4)

34:
(recharge dronea charging2 b4 b5)

35:
(fly dronea charging2 s3 b5 b1 b4)

36:
(deliver-supply dronea comms1 s3)

37:
```

```
(fly dronea s3 s4 b4 b1 b3)

38:
(inspect dronea s4)

39:
(fly dronea s4 s3 b3 b1 b2)

40:
(fly dronea s3 charging2 b2 b1 b1)

41:
(recharge dronea charging2 b1 b2)

42:
(recharge dronea charging2 b2 b3)

43:
(fly dronea charging2 charging3 b3 b2 b1)

44:
(recharge dronea charging3 b1 b2)

45:
(recharge dronea charging3 b2 b3)

46:
(recharge dronea charging3 b3 b4)

47:
(fly dronea charging3 depot2 b4 b1 b3)

48:
(pick-up-supply dronea medkit2 depot2)

49:
(fly dronea depot2 s5 b3 b1 b2)

50:
(deliver-supply dronea medkit2 s5)

51:
(fly dronea s5 s6 b2 b2 b0)

52:
(inspect dronea s6)


Plan Validation details
-----------------------
```

```
Checking next happening (time 1)
Deleting (supply-at water1 depot1)
Deleting (empty dronea)
Adding (carrying dronea water1)
Updating (total-cost) (0) by 1 increase

Checking next happening (time 2)
Deleting (at dronea depot1)
Deleting (battery dronea b6)
Adding (at dronea s2)
Adding (battery dronea b4)
Updating (total-cost) (1) by 2 increase

Checking next happening (time 3)
Deleting (carrying dronea water1)
Deleting (needs water1 s2)
Adding (empty dronea)
Adding (delivered water1 s2)
Updating (total-cost) (3) by 1 increase

Checking next happening (time 4)
Adding (inspected s2)
Updating (total-cost) (4) by 1 increase

Checking next happening (time 5)
Deleting (at dronea s2)
Deleting (battery dronea b4)
Adding (at dronea depot1)
Adding (battery dronea b2)
Updating (total-cost) (5) by 2 increase

Checking next happening (time 6)
Deleting (supply-at medkit1 depot1)
Deleting (empty dronea)
Adding (carrying dronea medkit1)
Updating (total-cost) (7) by 1 increase

Checking next happening (time 7)
Deleting (at dronea depot1)
Deleting (battery dronea b2)
Adding (at dronea charging1)
Adding (battery dronea b1)
Updating (total-cost) (8) by 1 increase

Checking next happening (time 8)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (9) by 1 increase

Checking next happening (time 9)
Deleting (battery dronea b2)
```

```
Adding (battery dronea b3)
Updating (total-cost) (10) by 1 increase

Checking next happening (time 10)
Deleting (battery dronea b3)
Adding (battery dronea b4)
Updating (total-cost) (11) by 1 increase

Checking next happening (time 11)
Deleting (battery dronea b4)
Adding (battery dronea b5)
Updating (total-cost) (12) by 1 increase

Checking next happening (time 12)
Deleting (at dronea charging1)
Deleting (battery dronea b5)
Adding (at dronea depot1)
Adding (battery dronea b4)
Updating (total-cost) (13) by 1 increase

Checking next happening (time 13)
Deleting (at dronea depot1)
Deleting (battery dronea b4)
Adding (at dronea s1)
Adding (battery dronea b3)
Updating (total-cost) (14) by 1 increase

Checking next happening (time 14)
Deleting (carrying dronea medkit1)
Deleting (needs medkit1 s1)
Adding (empty dronea)
Adding (delivered medkit1 s1)
Updating (total-cost) (15) by 1 increase

Checking next happening (time 15)
Deleting (at dronea s1)
Deleting (battery dronea b3)
Adding (at dronea depot1)
Adding (battery dronea b2)
Updating (total-cost) (16) by 1 increase

Checking next happening (time 16)
Deleting (at dronea depot1)
Deleting (battery dronea b2)
Adding (at dronea charging1)
Adding (battery dronea b1)
Updating (total-cost) (17) by 1 increase

Checking next happening (time 17)
Deleting (battery dronea b1)
Adding (battery dronea b2)
```

```
Updating (total-cost) (18) by 1 increase

Checking next happening (time 18)
Deleting (battery dronea b2)
Adding (battery dronea b3)
Updating (total-cost) (19) by 1 increase

Checking next happening (time 19)
Deleting (at dronea charging1)
Deleting (battery dronea b3)
Adding (at dronea charging2)
Adding (battery dronea b1)
Updating (total-cost) (20) by 2 increase

Checking next happening (time 20)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (22) by 1 increase

Checking next happening (time 21)
Deleting (battery dronea b2)
Adding (battery dronea b3)
Updating (total-cost) (23) by 1 increase

Checking next happening (time 22)
Deleting (battery dronea b3)
Adding (battery dronea b4)
Updating (total-cost) (24) by 1 increase

Checking next happening (time 23)
Deleting (battery dronea b4)
Adding (battery dronea b5)
Updating (total-cost) (25) by 1 increase

Checking next happening (time 24)
Deleting (battery dronea b5)
Adding (battery dronea b6)
Updating (total-cost) (26) by 1 increase

Checking next happening (time 25)
Deleting (at dronea charging2)
Deleting (battery dronea b6)
Adding (at dronea charging3)
Adding (battery dronea b4)
Updating (total-cost) (27) by 2 increase

Checking next happening (time 26)
Deleting (at dronea charging3)
Deleting (battery dronea b4)
Adding (at dronea depot2)
Adding (battery dronea b3)
```

```
Updating (total-cost) (29) by 1 increase

Checking next happening (time 27)
Deleting (supply-at comms1 depot2)
Deleting (empty dronea)
Adding (carrying dronea comms1)
Updating (total-cost) (30) by 1 increase

Checking next happening (time 28)
Deleting (at dronea depot2)
Deleting (battery dronea b3)
Adding (at dronea charging3)
Adding (battery dronea b2)
Updating (total-cost) (31) by 1 increase

Checking next happening (time 29)
Deleting (battery dronea b2)
Adding (battery dronea b3)
Updating (total-cost) (32) by 1 increase

Checking next happening (time 30)
Deleting (battery dronea b3)
Adding (battery dronea b4)
Updating (total-cost) (33) by 1 increase

Checking next happening (time 31)
Deleting (battery dronea b4)
Adding (battery dronea b5)
Updating (total-cost) (34) by 1 increase

Checking next happening (time 32)
Deleting (battery dronea b5)
Adding (battery dronea b6)
Updating (total-cost) (35) by 1 increase

Checking next happening (time 33)
Deleting (at dronea charging3)
Deleting (battery dronea b6)
Adding (at dronea charging2)
Adding (battery dronea b4)
Updating (total-cost) (36) by 2 increase

Checking next happening (time 34)
Deleting (battery dronea b4)
Adding (battery dronea b5)
Updating (total-cost) (38) by 1 increase

Checking next happening (time 35)
Deleting (at dronea charging2)
Deleting (battery dronea b5)
Adding (at dronea s3)
```

```
Adding (battery dronea b4)
Updating (total-cost) (39) by 1 increase

Checking next happening (time 36)
Deleting (carrying dronea comms1)
Deleting (needs comms1 s3)
Adding (empty dronea)
Adding (delivered comms1 s3)
Adding (accessible s4)
Updating (total-cost) (40) by 1 increase

Checking next happening (time 37)
Deleting (at dronea s3)
Deleting (battery dronea b4)
Adding (at dronea s4)
Adding (battery dronea b3)
Updating (total-cost) (41) by 1 increase

Checking next happening (time 38)
Adding (inspected s4)
Updating (total-cost) (42) by 1 increase

Checking next happening (time 39)
Deleting (at dronea s4)
Deleting (battery dronea b3)
Adding (at dronea s3)
Adding (battery dronea b2)
Updating (total-cost) (43) by 1 increase

Checking next happening (time 40)
Deleting (at dronea s3)
Deleting (battery dronea b2)
Adding (at dronea charging2)
Adding (battery dronea b1)
Updating (total-cost) (44) by 1 increase

Checking next happening (time 41)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (45) by 1 increase

Checking next happening (time 42)
Deleting (battery dronea b2)
Adding (battery dronea b3)
Updating (total-cost) (46) by 1 increase

Checking next happening (time 43)
Deleting (at dronea charging2)
Deleting (battery dronea b3)
Adding (at dronea charging3)
Adding (battery dronea b1)
```

```
Updating (total-cost) (47) by 2 increase

Checking next happening (time 44)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (49) by 1 increase

Checking next happening (time 45)
Deleting (battery dronea b2)
Adding (battery dronea b3)
Updating (total-cost) (50) by 1 increase

Checking next happening (time 46)
Deleting (battery dronea b3)
Adding (battery dronea b4)
Updating (total-cost) (51) by 1 increase

Checking next happening (time 47)
Deleting (at dronea charging3)
Deleting (battery dronea b4)
Adding (at dronea depot2)
Adding (battery dronea b3)
Updating (total-cost) (52) by 1 increase

Checking next happening (time 48)
Deleting (supply-at medkit2 depot2)
Deleting (empty dronea)
Adding (carrying dronea medkit2)
Updating (total-cost) (53) by 1 increase

Checking next happening (time 49)
Deleting (at dronea depot2)
Deleting (battery dronea b3)
Adding (at dronea s5)
Adding (battery dronea b2)
Updating (total-cost) (54) by 1 increase

Checking next happening (time 50)
Deleting (carrying dronea medkit2)
Deleting (needs medkit2 s5)
Adding (empty dronea)
Adding (delivered medkit2 s5)
Updating (total-cost) (55) by 1 increase

Checking next happening (time 51)
Deleting (at dronea s5)
Deleting (battery dronea b2)
Adding (at dronea s6)
Adding (battery dronea b0)
Updating (total-cost) (56) by 2 increase
```

```
Checking next happening (time 52)
Adding (inspected s6)
Updating (total-cost) (58) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 59

Successful plans:
Value: 59
 plans/large-1drone-B-astar 59
```

A4.8 large1-B-lama

None

```
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/large-1drone-B-lama
Plan to validate:

Plan size: 84
1:
(pick-up-supply dronea water1 depot1)

2:
(fly dronea depot1 s2 b6 b2 b4)

3:
(deliver-supply dronea water1 s2)

4:
(inspect dronea s2)

5:
(fly dronea s2 depot1 b4 b2 b2)

6:
(fly dronea depot1 charging1 b2 b1 b1)

7:
(recharge dronea charging1 b1 b2)
```

```
8:
(fly dronea charging1 charging2 b2 b2 b0)

9:
(recharge dronea charging2 b0 b1)

10:
(recharge dronea charging2 b1 b2)

11:
(fly dronea charging2 charging3 b2 b2 b0)

12:
(recharge dronea charging3 b0 b1)

13:
(recharge dronea charging3 b1 b2)

14:
(fly dronea charging3 depot2 b2 b1 b1)

15:
(pick-up-supply dronea comms1 depot2)

16:
(fly dronea depot2 charging3 b1 b1 b0)

17:
(recharge dronea charging3 b0 b1)

18:
(recharge dronea charging3 b1 b2)

19:
(fly dronea charging3 charging2 b2 b2 b0)

20:
(recharge dronea charging2 b0 b1)

21:
(recharge dronea charging2 b1 b2)

22:
(fly dronea charging2 s3 b2 b1 b1)

23:
(deliver-supply dronea comms1 s3)

24:
(fly dronea s3 charging2 b1 b1 b0)
```

```
25:
(recharge dronea charging2 b0 b1)

26:
(recharge dronea charging2 b1 b2)

27:
(fly dronea charging2 charging1 b2 b2 b0)

28:
(recharge dronea charging1 b0 b1)

29:
(recharge dronea charging1 b1 b2)

30:
(fly dronea charging1 depot1 b2 b1 b1)

31:
(pick-up-supply dronea medkit1 depot1)

32:
(fly dronea depot1 charging1 b1 b1 b0)

33:
(recharge dronea charging1 b0 b1)

34:
(recharge dronea charging1 b1 b2)

35:
(recharge dronea charging1 b2 b3)

36:
(recharge dronea charging1 b3 b4)

37:
(fly dronea charging1 depot1 b4 b1 b3)

38:
(fly dronea depot1 s1 b3 b1 b2)

39:
(deliver-supply dronea medkit1 s1)

40:
(fly dronea s1 depot1 b2 b1 b1)

41:
(fly dronea depot1 charging1 b1 b1 b0)
```

```
42:
(recharge dronea charging1 b0 b1)

43:
(recharge dronea charging1 b1 b2)

44:
(fly dronea charging1 charging2 b2 b2 b0)

45:
(recharge dronea charging2 b0 b1)

46:
(recharge dronea charging2 b1 b2)

47:
(fly dronea charging2 charging3 b2 b2 b0)

48:
(recharge dronea charging3 b0 b1)

49:
(recharge dronea charging3 b1 b2)

50:
(fly dronea charging3 depot2 b2 b1 b1)

51:
(pick-up-supply dronea medkit2 depot2)

52:
(fly dronea depot2 charging3 b1 b1 b0)

53:
(recharge dronea charging3 b0 b1)

54:
(recharge dronea charging3 b1 b2)

55:
(recharge dronea charging3 b2 b3)

56:
(recharge dronea charging3 b3 b4)

57:
(fly dronea charging3 depot2 b4 b1 b3)

58:
(fly dronea depot2 s5 b3 b1 b2)
```

```
59:
(deliver-supply dronea medkit2 s5)

60:
(fly dronea s5 depot2 b2 b1 b1)

61:
(fly dronea depot2 charging3 b1 b1 b0)

62:
(recharge dronea charging3 b0 b1)

63:
(recharge dronea charging3 b1 b2)

64:
(fly dronea charging3 charging2 b2 b2 b0)

65:
(recharge dronea charging2 b0 b1)

66:
(recharge dronea charging2 b1 b2)

67:
(recharge dronea charging2 b2 b3)

68:
(recharge dronea charging2 b3 b4)

69:
(fly dronea charging2 s3 b4 b1 b3)

70:
(fly dronea s3 s4 b3 b1 b2)

71:
(inspect dronea s4)

72:
(fly dronea s4 s3 b2 b1 b1)

73:
(fly dronea s3 charging2 b1 b1 b0)

74:
(recharge dronea charging2 b0 b1)

75:
(recharge dronea charging2 b1 b2)
```

```
76:
(fly dronea charging2 charging3 b2 b2 b0)

77:
(recharge dronea charging3 b0 b1)

78:
(recharge dronea charging3 b1 b2)

79:
(recharge dronea charging3 b2 b3)

80:
(recharge dronea charging3 b3 b4)

81:
(fly dronea charging3 depot2 b4 b1 b3)

82:
(fly dronea depot2 s5 b3 b1 b2)

83:
(fly dronea s5 s6 b2 b2 b0)

84:
(inspect dronea s6)


Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (supply-at water1 depot1)
Deleting (empty dronea)
Adding (carrying dronea water1)
Updating (total-cost) (0) by 1 increase

Checking next happening (time 2)
Deleting (at dronea depot1)
Deleting (battery dronea b6)
Adding (at dronea s2)
Adding (battery dronea b4)
Updating (total-cost) (1) by 2 increase

Checking next happening (time 3)
Deleting (carrying dronea water1)
Deleting (needs water1 s2)
Adding (empty dronea)
Adding (delivered water1 s2)
Updating (total-cost) (3) by 1 increase
```

```
Checking next happening (time 4)
Adding (inspected s2)
Updating (total-cost) (4) by 1 increase

Checking next happening (time 5)
Deleting (at dronea s2)
Deleting (battery dronea b4)
Adding (at dronea depot1)
Adding (battery dronea b2)
Updating (total-cost) (5) by 2 increase

Checking next happening (time 6)
Deleting (at dronea depot1)
Deleting (battery dronea b2)
Adding (at dronea charging1)
Adding (battery dronea b1)
Updating (total-cost) (7) by 1 increase

Checking next happening (time 7)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (8) by 1 increase

Checking next happening (time 8)
Deleting (at dronea charging1)
Deleting (battery dronea b2)
Adding (at dronea charging2)
Adding (battery dronea b0)
Updating (total-cost) (9) by 2 increase

Checking next happening (time 9)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (11) by 1 increase

Checking next happening (time 10)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (12) by 1 increase

Checking next happening (time 11)
Deleting (at dronea charging2)
Deleting (battery dronea b2)
Adding (at dronea charging3)
Adding (battery dronea b0)
Updating (total-cost) (13) by 2 increase

Checking next happening (time 12)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (15) by 1 increase
```

```
Checking next happening (time 13)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (16) by 1 increase

Checking next happening (time 14)
Deleting (at dronea charging3)
Deleting (battery dronea b2)
Adding (at dronea depot2)
Adding (battery dronea b1)
Updating (total-cost) (17) by 1 increase

Checking next happening (time 15)
Deleting (supply-at comms1 depot2)
Deleting (empty dronea)
Adding (carrying dronea comms1)
Updating (total-cost) (18) by 1 increase

Checking next happening (time 16)
Deleting (at dronea depot2)
Deleting (battery dronea b1)
Adding (at dronea charging3)
Adding (battery dronea b0)
Updating (total-cost) (19) by 1 increase

Checking next happening (time 17)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (20) by 1 increase

Checking next happening (time 18)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (21) by 1 increase

Checking next happening (time 19)
Deleting (at dronea charging3)
Deleting (battery dronea b2)
Adding (at dronea charging2)
Adding (battery dronea b0)
Updating (total-cost) (22) by 2 increase

Checking next happening (time 20)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (24) by 1 increase

Checking next happening (time 21)
Deleting (battery dronea b1)
Adding (battery dronea b2)
```

```
Updating (total-cost) (25) by 1 increase

Checking next happening (time 22)
Deleting (at dronea charging2)
Deleting (battery dronea b2)
Adding (at dronea s3)
Adding (battery dronea b1)
Updating (total-cost) (26) by 1 increase

Checking next happening (time 23)
Deleting (carrying dronea comms1)
Deleting (needs comms1 s3)
Adding (empty dronea)
Adding (delivered comms1 s3)
Adding (accessible s4)
Updating (total-cost) (27) by 1 increase

Checking next happening (time 24)
Deleting (at dronea s3)
Deleting (battery dronea b1)
Adding (at dronea charging2)
Adding (battery dronea b0)
Updating (total-cost) (28) by 1 increase

Checking next happening (time 25)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (29) by 1 increase

Checking next happening (time 26)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (30) by 1 increase

Checking next happening (time 27)
Deleting (at dronea charging2)
Deleting (battery dronea b2)
Adding (at dronea charging1)
Adding (battery dronea b0)
Updating (total-cost) (31) by 2 increase

Checking next happening (time 28)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (33) by 1 increase

Checking next happening (time 29)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (34) by 1 increase
```

```
Checking next happening (time 30)
Deleting (at dronea charging1)
Deleting (battery dronea b2)
Adding (at dronea depot1)
Adding (battery dronea b1)
Updating (total-cost) (35) by 1 increase

Checking next happening (time 31)
Deleting (supply-at medkit1 depot1)
Deleting (empty dronea)
Adding (carrying dronea medkit1)
Updating (total-cost) (36) by 1 increase

Checking next happening (time 32)
Deleting (at dronea depot1)
Deleting (battery dronea b1)
Adding (at dronea charging1)
Adding (battery dronea b0)
Updating (total-cost) (37) by 1 increase

Checking next happening (time 33)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (38) by 1 increase

Checking next happening (time 34)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (39) by 1 increase

Checking next happening (time 35)
Deleting (battery dronea b2)
Adding (battery dronea b3)
Updating (total-cost) (40) by 1 increase

Checking next happening (time 36)
Deleting (battery dronea b3)
Adding (battery dronea b4)
Updating (total-cost) (41) by 1 increase

Checking next happening (time 37)
Deleting (at dronea charging1)
Deleting (battery dronea b4)
Adding (at dronea depot1)
Adding (battery dronea b3)
Updating (total-cost) (42) by 1 increase

Checking next happening (time 38)
Deleting (at dronea depot1)
Deleting (battery dronea b3)
Adding (at dronea s1)
```

```
Adding (battery dronea b2)
Updating (total-cost) (43) by 1 increase

Checking next happening (time 39)
Deleting (carrying dronea medkit1)
Deleting (needs medkit1 s1)
Adding (empty dronea)
Adding (delivered medkit1 s1)
Updating (total-cost) (44) by 1 increase

Checking next happening (time 40)
Deleting (at dronea s1)
Deleting (battery dronea b2)
Adding (at dronea depot1)
Adding (battery dronea b1)
Updating (total-cost) (45) by 1 increase

Checking next happening (time 41)
Deleting (at dronea depot1)
Deleting (battery dronea b1)
Adding (at dronea charging1)
Adding (battery dronea b0)
Updating (total-cost) (46) by 1 increase

Checking next happening (time 42)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (47) by 1 increase

Checking next happening (time 43)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (48) by 1 increase

Checking next happening (time 44)
Deleting (at dronea charging1)
Deleting (battery dronea b2)
Adding (at dronea charging2)
Adding (battery dronea b0)
Updating (total-cost) (49) by 2 increase

Checking next happening (time 45)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (51) by 1 increase

Checking next happening (time 46)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (52) by 1 increase
```

```
Checking next happening (time 47)
Deleting (at dronea charging2)
Deleting (battery dronea b2)
Adding (at dronea charging3)
Adding (battery dronea b0)
Updating (total-cost) (53) by 2 increase

Checking next happening (time 48)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (55) by 1 increase

Checking next happening (time 49)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (56) by 1 increase

Checking next happening (time 50)
Deleting (at dronea charging3)
Deleting (battery dronea b2)
Adding (at dronea depot2)
Adding (battery dronea b1)
Updating (total-cost) (57) by 1 increase

Checking next happening (time 51)
Deleting (supply-at medkit2 depot2)
Deleting (empty dronea)
Adding (carrying dronea medkit2)
Updating (total-cost) (58) by 1 increase

Checking next happening (time 52)
Deleting (at dronea depot2)
Deleting (battery dronea b1)
Adding (at dronea charging3)
Adding (battery dronea b0)
Updating (total-cost) (59) by 1 increase

Checking next happening (time 53)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (60) by 1 increase

Checking next happening (time 54)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (61) by 1 increase

Checking next happening (time 55)
Deleting (battery dronea b2)
Adding (battery dronea b3)
Updating (total-cost) (62) by 1 increase
```

```
Checking next happening (time 56)
Deleting (battery dronea b3)
Adding (battery dronea b4)
Updating (total-cost) (63) by 1 increase

Checking next happening (time 57)
Deleting (at dronea charging3)
Deleting (battery dronea b4)
Adding (at dronea depot2)
Adding (battery dronea b3)
Updating (total-cost) (64) by 1 increase

Checking next happening (time 58)
Deleting (at dronea depot2)
Deleting (battery dronea b3)
Adding (at dronea s5)
Adding (battery dronea b2)
Updating (total-cost) (65) by 1 increase

Checking next happening (time 59)
Deleting (carrying dronea medkit2)
Deleting (needs medkit2 s5)
Adding (empty dronea)
Adding (delivered medkit2 s5)
Updating (total-cost) (66) by 1 increase

Checking next happening (time 60)
Deleting (at dronea s5)
Deleting (battery dronea b2)
Adding (at dronea depot2)
Adding (battery dronea b1)
Updating (total-cost) (67) by 1 increase

Checking next happening (time 61)
Deleting (at dronea depot2)
Deleting (battery dronea b1)
Adding (at dronea charging3)
Adding (battery dronea b0)
Updating (total-cost) (68) by 1 increase

Checking next happening (time 62)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (69) by 1 increase

Checking next happening (time 63)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (70) by 1 increase
```

```
Checking next happening (time 64)
Deleting (at dronea charging3)
Deleting (battery dronea b2)
Adding (at dronea charging2)
Adding (battery dronea b0)
Updating (total-cost) (71) by 2 increase

Checking next happening (time 65)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (73) by 1 increase

Checking next happening (time 66)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (74) by 1 increase

Checking next happening (time 67)
Deleting (battery dronea b2)
Adding (battery dronea b3)
Updating (total-cost) (75) by 1 increase

Checking next happening (time 68)
Deleting (battery dronea b3)
Adding (battery dronea b4)
Updating (total-cost) (76) by 1 increase

Checking next happening (time 69)
Deleting (at dronea charging2)
Deleting (battery dronea b4)
Adding (at dronea s3)
Adding (battery dronea b3)
Updating (total-cost) (77) by 1 increase

Checking next happening (time 70)
Deleting (at dronea s3)
Deleting (battery dronea b3)
Adding (at dronea s4)
Adding (battery dronea b2)
Updating (total-cost) (78) by 1 increase

Checking next happening (time 71)
Adding (inspected s4)
Updating (total-cost) (79) by 1 increase

Checking next happening (time 72)
Deleting (at dronea s4)
Deleting (battery dronea b2)
Adding (at dronea s3)
Adding (battery dronea b1)
Updating (total-cost) (80) by 1 increase
```

```
Checking next happening (time 73)
Deleting (at dronea s3)
Deleting (battery dronea b1)
Adding (at dronea charging2)
Adding (battery dronea b0)
Updating (total-cost) (81) by 1 increase

Checking next happening (time 74)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (82) by 1 increase

Checking next happening (time 75)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (83) by 1 increase

Checking next happening (time 76)
Deleting (at dronea charging2)
Deleting (battery dronea b2)
Adding (at dronea charging3)
Adding (battery dronea b0)
Updating (total-cost) (84) by 2 increase

Checking next happening (time 77)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (86) by 1 increase

Checking next happening (time 78)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (87) by 1 increase

Checking next happening (time 79)
Deleting (battery dronea b2)
Adding (battery dronea b3)
Updating (total-cost) (88) by 1 increase

Checking next happening (time 80)
Deleting (battery dronea b3)
Adding (battery dronea b4)
Updating (total-cost) (89) by 1 increase

Checking next happening (time 81)
Deleting (at dronea charging3)
Deleting (battery dronea b4)
Adding (at dronea depot2)
Adding (battery dronea b3)
Updating (total-cost) (90) by 1 increase
```

```
Checking next happening (time 82)
Deleting (at dronea depot2)
Deleting (battery dronea b3)
Adding (at dronea s5)
Adding (battery dronea b2)
Updating (total-cost) (91) by 1 increase

Checking next happening (time 83)
Deleting (at dronea s5)
Deleting (battery dronea b2)
Adding (at dronea s6)
Adding (battery dronea b0)
Updating (total-cost) (92) by 2 increase

Checking next happening (time 84)
Adding (inspected s6)
Updating (total-cost) (94) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 95

Successful plans:
Value: 95
 plans/large-1drone-B-lama 95
```

A4.9 medium-A-astar

None

```
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/medium-A-astar
Plan to validate:

Plan size: 21
1:
(pick-up-supply dronea medkit1 depot)

2:
(fly droneb charging2 s3 b6 b1 b5)
```

```
3:
(inspect droneb s3)

4:
(fly droneb s3 depot b5 b2 b3)

5:
(pick-up-supply droneb water1 depot)

6:
(fly droneb depot s3 b3 b2 b1)

7:
(fly droneb s3 charging2 b1 b1 b0)

8:
(recharge droneb charging2 b0 b6)

9:
(fly droneb charging2 s4 b6 b2 b4)

10:
(inspect droneb s4)

11:
(fly droneb s4 s5 b4 b1 b3)

12:
(inspect droneb s5)

13:
(deliver-supply droneb water1 s5)

14:
(fly dronea depot s1 b6 b1 b5)

15:
(deliver-supply dronea medkit1 s1)

16:
(fly dronea s1 depot b5 b1 b4)

17:
(pick-up-supply dronea medkit2 depot)

18:
(fly dronea depot s1 b4 b1 b3)

19:
(fly dronea s1 s2 b3 b2 b1)
```

```
20:
(inspect dronea s2)

21:
(deliver-supply dronea medkit2 s2)


Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (supply-at medkit1 depot)
Deleting (empty dronea)
Adding (carrying dronea medkit1)
Updating (total-cost) (0) by 1 increase

Checking next happening (time 2)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
Adding (at droneb s3)
Adding (battery droneb b5)
Updating (total-cost) (1) by 1 increase

Checking next happening (time 3)
Adding (inspected s3)
Updating (total-cost) (2) by 1 increase

Checking next happening (time 4)
Deleting (at droneb s3)
Deleting (battery droneb b5)
Adding (at droneb depot)
Adding (battery droneb b3)
Updating (total-cost) (3) by 2 increase

Checking next happening (time 5)
Deleting (supply-at water1 depot)
Deleting (empty droneb)
Adding (carrying droneb water1)
Updating (total-cost) (5) by 1 increase

Checking next happening (time 6)
Deleting (at droneb depot)
Deleting (battery droneb b3)
Adding (at droneb s3)
Adding (battery droneb b1)
Updating (total-cost) (6) by 2 increase

Checking next happening (time 7)
Deleting (at droneb s3)
Deleting (battery droneb b1)
Adding (at droneb charging2)
```

```
Adding (battery droneb b0)
Updating (total-cost) (8) by 1 increase

Checking next happening (time 8)
Deleting (battery droneb b0)
Adding (battery droneb b6)
Updating (total-cost) (9) by 1 increase

Checking next happening (time 9)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
Adding (at droneb s4)
Adding (battery droneb b4)
Updating (total-cost) (10) by 2 increase

Checking next happening (time 10)
Adding (inspected s4)
Adding (accessible s5)
Updating (total-cost) (12) by 1 increase

Checking next happening (time 11)
Deleting (at droneb s4)
Deleting (battery droneb b4)
Adding (at droneb s5)
Adding (battery droneb b3)
Updating (total-cost) (13) by 1 increase

Checking next happening (time 12)
Adding (inspected s5)
Updating (total-cost) (14) by 1 increase

Checking next happening (time 13)
Deleting (carrying droneb water1)
Deleting (needs water1 s5)
Adding (empty droneb)
Adding (delivered water1 s5)
Updating (total-cost) (15) by 1 increase

Checking next happening (time 14)
Deleting (at dronea depot)
Deleting (battery dronea b6)
Adding (at dronea s1)
Adding (battery dronea b5)
Updating (total-cost) (16) by 1 increase

Checking next happening (time 15)
Deleting (carrying dronea medkit1)
Deleting (needs medkit1 s1)
Adding (empty dronea)
Adding (delivered medkit1 s1)
Updating (total-cost) (17) by 1 increase
```

```
Checking next happening (time 16)
Deleting (at dronea s1)
Deleting (battery dronea b5)
Adding (at dronea depot)
Adding (battery dronea b4)
Updating (total-cost) (18) by 1 increase

Checking next happening (time 17)
Deleting (supply-at medkit2 depot)
Deleting (empty dronea)
Adding (carrying dronea medkit2)
Updating (total-cost) (19) by 1 increase

Checking next happening (time 18)
Deleting (at dronea depot)
Deleting (battery dronea b4)
Adding (at dronea s1)
Adding (battery dronea b3)
Updating (total-cost) (20) by 1 increase

Checking next happening (time 19)
Deleting (at dronea s1)
Deleting (battery dronea b3)
Adding (at dronea s2)
Adding (battery dronea b1)
Updating (total-cost) (21) by 2 increase

Checking next happening (time 20)
Adding (inspected s2)
Updating (total-cost) (23) by 1 increase

Checking next happening (time 21)
Deleting (carrying dronea medkit2)
Deleting (needs medkit2 s2)
Adding (empty dronea)
Adding (delivered medkit2 s2)
Updating (total-cost) (24) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 25

Successful plans:
Value: 25
 plans/medium-A-astar 25
```

A4.10 medium-A-lama

None

```
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/medium-A-lama
Plan to validate:

Plan size: 38
1:
(fly dronea depot s1 b6 b1 b5)

2:
(fly droneb charging2 s3 b6 b1 b5)

3:
(inspect droneb s3)

4:
(fly dronea s1 depot b5 b1 b4)

5:
(pick-up-supply dronea medkit1 depot)

6:
(fly dronea depot s1 b4 b1 b3)

7:
(deliver-supply dronea medkit1 s1)

8:
(fly dronea s1 s2 b3 b2 b1)

9:
(inspect dronea s2)

10:
(fly droneb s3 charging2 b5 b1 b4)

11:
(fly droneb charging2 s4 b4 b2 b2)

12:
(inspect droneb s4)
```

```
13:
(fly droneb s4 charging2 b2 b2 b0)

14:
(recharge droneb charging2 b0 b6)

15:
(fly droneb charging2 s4 b6 b2 b4)

16:
(fly droneb s4 s5 b4 b1 b3)

17:
(inspect droneb s5)

18:
(fly droneb s5 s4 b3 b1 b2)

19:
(fly droneb s4 charging2 b2 b2 b0)

20:
(recharge droneb charging2 b0 b6)

21:
(fly droneb charging2 s3 b6 b1 b5)

22:
(fly droneb s3 depot b5 b2 b3)

23:
(pick-up-supply droneb water1 depot)

24:
(fly droneb depot s3 b3 b2 b1)

25:
(fly droneb s3 charging2 b1 b1 b0)

26:
(recharge droneb charging2 b0 b6)

27:
(fly droneb charging2 s4 b6 b2 b4)

28:
(fly droneb s4 s5 b4 b1 b3)

29:
(deliver-supply droneb water1 s5)
```

```
30:
(fly droneb s5 s4 b3 b1 b2)

31:
(fly droneb s4 charging2 b2 b2 b0)

32:
(recharge droneb charging2 b0 b6)

33:
(fly droneb charging2 s3 b6 b1 b5)

34:
(fly droneb s3 depot b5 b2 b3)

35:
(pick-up-supply droneb medkit2 depot)

36:
(fly droneb depot s1 b3 b1 b2)

37:
(fly droneb s1 s2 b2 b2 b0)

38:
(deliver-supply droneb medkit2 s2)


Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (at dronea depot)
Deleting (battery dronea b6)
Adding (at dronea s1)
Adding (battery dronea b5)
Updating (total-cost) (0) by 1 increase

Checking next happening (time 2)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
Adding (at droneb s3)
Adding (battery droneb b5)
Updating (total-cost) (1) by 1 increase

Checking next happening (time 3)
Adding (inspected s3)
Updating (total-cost) (2) by 1 increase

Checking next happening (time 4)
```

```
Deleting (at dronea s1)
Deleting (battery dronea b5)
Adding (at dronea depot)
Adding (battery dronea b4)
Updating (total-cost) (3) by 1 increase

Checking next happening (time 5)
Deleting (supply-at medkit1 depot)
Deleting (empty dronea)
Adding (carrying dronea medkit1)
Updating (total-cost) (4) by 1 increase

Checking next happening (time 6)
Deleting (at dronea depot)
Deleting (battery dronea b4)
Adding (at dronea s1)
Adding (battery dronea b3)
Updating (total-cost) (5) by 1 increase

Checking next happening (time 7)
Deleting (carrying dronea medkit1)
Deleting (needs medkit1 s1)
Adding (empty dronea)
Adding (delivered medkit1 s1)
Updating (total-cost) (6) by 1 increase

Checking next happening (time 8)
Deleting (at dronea s1)
Deleting (battery dronea b3)
Adding (at dronea s2)
Adding (battery dronea b1)
Updating (total-cost) (7) by 2 increase

Checking next happening (time 9)
Adding (inspected s2)
Updating (total-cost) (9) by 1 increase

Checking next happening (time 10)
Deleting (at droneb s3)
Deleting (battery droneb b5)
Adding (at droneb charging2)
Adding (battery droneb b4)
Updating (total-cost) (10) by 1 increase

Checking next happening (time 11)
Deleting (at droneb charging2)
Deleting (battery droneb b4)
Adding (at droneb s4)
Adding (battery droneb b2)
Updating (total-cost) (11) by 2 increase
```

```
Checking next happening (time 12)
Adding (inspected s4)
Adding (accessible s5)
Updating (total-cost) (13) by 1 increase

Checking next happening (time 13)
Deleting (at droneb s4)
Deleting (battery droneb b2)
Adding (at droneb charging2)
Adding (battery droneb b0)
Updating (total-cost) (14) by 2 increase

Checking next happening (time 14)
Deleting (battery droneb b0)
Adding (battery droneb b6)
Updating (total-cost) (16) by 1 increase

Checking next happening (time 15)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
Adding (at droneb s4)
Adding (battery droneb b4)
Updating (total-cost) (17) by 2 increase

Checking next happening (time 16)
Deleting (at droneb s4)
Deleting (battery droneb b4)
Adding (at droneb s5)
Adding (battery droneb b3)
Updating (total-cost) (19) by 1 increase

Checking next happening (time 17)
Adding (inspected s5)
Updating (total-cost) (20) by 1 increase

Checking next happening (time 18)
Deleting (at droneb s5)
Deleting (battery droneb b3)
Adding (at droneb s4)
Adding (battery droneb b2)
Updating (total-cost) (21) by 1 increase

Checking next happening (time 19)
Deleting (at droneb s4)
Deleting (battery droneb b2)
Adding (at droneb charging2)
Adding (battery droneb b0)
Updating (total-cost) (22) by 2 increase

Checking next happening (time 20)
Deleting (battery droneb b0)
```

```
Adding (battery droneb b6)
Updating (total-cost) (24) by 1 increase

Checking next happening (time 21)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
Adding (at droneb s3)
Adding (battery droneb b5)
Updating (total-cost) (25) by 1 increase

Checking next happening (time 22)
Deleting (at droneb s3)
Deleting (battery droneb b5)
Adding (at droneb depot)
Adding (battery droneb b3)
Updating (total-cost) (26) by 2 increase

Checking next happening (time 23)
Deleting (supply-at water1 depot)
Deleting (empty droneb)
Adding (carrying droneb water1)
Updating (total-cost) (28) by 1 increase

Checking next happening (time 24)
Deleting (at droneb depot)
Deleting (battery droneb b3)
Adding (at droneb s3)
Adding (battery droneb b1)
Updating (total-cost) (29) by 2 increase

Checking next happening (time 25)
Deleting (at droneb s3)
Deleting (battery droneb b1)
Adding (at droneb charging2)
Adding (battery droneb b0)
Updating (total-cost) (31) by 1 increase

Checking next happening (time 26)
Deleting (battery droneb b0)
Adding (battery droneb b6)
Updating (total-cost) (32) by 1 increase

Checking next happening (time 27)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
Adding (at droneb s4)
Adding (battery droneb b4)
Updating (total-cost) (33) by 2 increase

Checking next happening (time 28)
Deleting (at droneb s4)
```

```
Deleting (battery droneb b4)
Adding (at droneb s5)
Adding (battery droneb b3)
Updating (total-cost) (35) by 1 increase

Checking next happening (time 29)
Deleting (carrying droneb water1)
Deleting (needs water1 s5)
Adding (empty droneb)
Adding (delivered water1 s5)
Updating (total-cost) (36) by 1 increase

Checking next happening (time 30)
Deleting (at droneb s5)
Deleting (battery droneb b3)
Adding (at droneb s4)
Adding (battery droneb b2)
Updating (total-cost) (37) by 1 increase

Checking next happening (time 31)
Deleting (at droneb s4)
Deleting (battery droneb b2)
Adding (at droneb charging2)
Adding (battery droneb b0)
Updating (total-cost) (38) by 2 increase

Checking next happening (time 32)
Deleting (battery droneb b0)
Adding (battery droneb b6)
Updating (total-cost) (40) by 1 increase

Checking next happening (time 33)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
Adding (at droneb s3)
Adding (battery droneb b5)
Updating (total-cost) (41) by 1 increase

Checking next happening (time 34)
Deleting (at droneb s3)
Deleting (battery droneb b5)
Adding (at droneb depot)
Adding (battery droneb b3)
Updating (total-cost) (42) by 2 increase

Checking next happening (time 35)
Deleting (supply-at medkit2 depot)
Deleting (empty droneb)
Adding (carrying droneb medkit2)
Updating (total-cost) (44) by 1 increase
```

```
Checking next happening (time 36)
Deleting (at droneb depot)
Deleting (battery droneb b3)
Adding (at droneb s1)
Adding (battery droneb b2)
Updating (total-cost) (45) by 1 increase

Checking next happening (time 37)
Deleting (at droneb s1)
Deleting (battery droneb b2)
Adding (at droneb s2)
Adding (battery droneb b0)
Updating (total-cost) (46) by 2 increase

Checking next happening (time 38)
Deleting (carrying droneb medkit2)
Deleting (needs medkit2 s2)
Adding (empty droneb)
Adding (delivered medkit2 s2)
Updating (total-cost) (48) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 49

Successful plans:
Value: 49
 plans/medium-A-lama 49
```

A4.11 medium-B-astar

None

```
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/medium-B-astar
Plan to validate:

Plan size: 22
1:
(pick-up-supply dronea medkit1 depot)

2:
```

```
(fly droneb charging2 s3 b6 b1 b5)

3:
(inspect droneb s3)

4:
(fly droneb s3 depot b5 b2 b3)

5:
(pick-up-supply droneb medkit2 depot)

6:
(fly droneb depot s1 b3 b1 b2)

7:
(fly droneb s1 s2 b2 b2 b0)

8:
(inspect droneb s2)

9:
(deliver-supply droneb medkit2 s2)

10:
(fly dronea depot s1 b6 b1 b5)

11:
(deliver-supply dronea medkit1 s1)

12:
(fly dronea s1 depot b5 b1 b4)

13:
(pick-up-supply dronea water1 depot)

14:
(fly dronea depot s3 b4 b2 b2)

15:
(fly dronea s3 charging2 b2 b1 b1)

16:
(recharge dronea charging2 b1 b2)

17:
(recharge dronea charging2 b2 b3)

18:
(fly dronea charging2 s4 b3 b2 b1)

19:
```

```
(inspect dronea s4)

20:
(fly dronea s4 s5 b1 b1 b0)

21:
(inspect dronea s5)

22:
(deliver-supply dronea water1 s5)


Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (supply-at medkit1 depot)
Deleting (empty dronea)
Adding (carrying dronea medkit1)
Updating (total-cost) (0) by 1 increase

Checking next happening (time 2)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
Adding (at droneb s3)
Adding (battery droneb b5)
Updating (total-cost) (1) by 1 increase

Checking next happening (time 3)
Adding (inspected s3)
Updating (total-cost) (2) by 1 increase

Checking next happening (time 4)
Deleting (at droneb s3)
Deleting (battery droneb b5)
Adding (at droneb depot)
Adding (battery droneb b3)
Updating (total-cost) (3) by 2 increase

Checking next happening (time 5)
Deleting (supply-at medkit2 depot)
Deleting (empty droneb)
Adding (carrying droneb medkit2)
Updating (total-cost) (5) by 1 increase

Checking next happening (time 6)
Deleting (at droneb depot)
Deleting (battery droneb b3)
Adding (at droneb s1)
Adding (battery droneb b2)
Updating (total-cost) (6) by 1 increase
```

```
Checking next happening (time 7)
Deleting (at droneb s1)
Deleting (battery droneb b2)
Adding (at droneb s2)
Adding (battery droneb b0)
Updating (total-cost) (7) by 2 increase

Checking next happening (time 8)
Adding (inspected s2)
Updating (total-cost) (9) by 1 increase

Checking next happening (time 9)
Deleting (carrying droneb medkit2)
Deleting (needs medkit2 s2)
Adding (empty droneb)
Adding (delivered medkit2 s2)
Updating (total-cost) (10) by 1 increase

Checking next happening (time 10)
Deleting (at dronea depot)
Deleting (battery dronea b6)
Adding (at dronea s1)
Adding (battery dronea b5)
Updating (total-cost) (11) by 1 increase

Checking next happening (time 11)
Deleting (carrying dronea medkit1)
Deleting (needs medkit1 s1)
Adding (empty dronea)
Adding (delivered medkit1 s1)
Updating (total-cost) (12) by 1 increase

Checking next happening (time 12)
Deleting (at dronea s1)
Deleting (battery dronea b5)
Adding (at dronea depot)
Adding (battery dronea b4)
Updating (total-cost) (13) by 1 increase

Checking next happening (time 13)
Deleting (supply-at water1 depot)
Deleting (empty dronea)
Adding (carrying dronea water1)
Updating (total-cost) (14) by 1 increase

Checking next happening (time 14)
Deleting (at dronea depot)
Deleting (battery dronea b4)
Adding (at dronea s3)
Adding (battery dronea b2)
```

```
Updating (total-cost) (15) by 2 increase

Checking next happening (time 15)
Deleting (at dronea s3)
Deleting (battery dronea b2)
Adding (at dronea charging2)
Adding (battery dronea b1)
Updating (total-cost) (17) by 1 increase

Checking next happening (time 16)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (18) by 1 increase

Checking next happening (time 17)
Deleting (battery dronea b2)
Adding (battery dronea b3)
Updating (total-cost) (19) by 1 increase

Checking next happening (time 18)
Deleting (at dronea charging2)
Deleting (battery dronea b3)
Adding (at dronea s4)
Adding (battery dronea b1)
Updating (total-cost) (20) by 2 increase

Checking next happening (time 19)
Adding (inspected s4)
Adding (accessible s5)
Updating (total-cost) (22) by 1 increase

Checking next happening (time 20)
Deleting (at dronea s4)
Deleting (battery dronea b1)
Adding (at dronea s5)
Adding (battery dronea b0)
Updating (total-cost) (23) by 1 increase

Checking next happening (time 21)
Adding (inspected s5)
Updating (total-cost) (24) by 1 increase

Checking next happening (time 22)
Deleting (carrying dronea water1)
Deleting (needs water1 s5)
Adding (empty dronea)
Adding (delivered water1 s5)
Updating (total-cost) (25) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 26
```

```
Successful plans:
Value: 26
 plans/medium-B-astar 26
```

A4.12 medium-B-lama

None

```
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/medium-B-lama
Plan to validate:

Plan size: 23
1:
(pick-up-supply dronea medkit1 depot)

2:
(fly dronea depot s1 b6 b1 b5)

3:
(deliver-supply dronea medkit1 s1)

4:
(fly droneb charging2 s3 b6 b1 b5)

5:
(inspect droneb s3)

6:
(fly dronea s1 depot b5 b1 b4)

7:
(pick-up-supply dronea medkit2 depot)

8:
(fly dronea depot s1 b4 b1 b3)

9:
(fly dronea s1 s2 b3 b2 b1)

10:
```

```
(inspect dronea s2)

11:
(deliver-supply dronea medkit2 s2)

12:
(fly droneb s3 depot b5 b2 b3)

13:
(pick-up-supply droneb water1 depot)

14:
(fly droneb depot s3 b3 b2 b1)

15:
(fly droneb s3 charging2 b1 b1 b0)

16:
(recharge droneb charging2 b0 b1)

17:
(recharge droneb charging2 b1 b2)

18:
(recharge droneb charging2 b2 b3)

19:
(fly droneb charging2 s4 b3 b2 b1)

20:
(inspect droneb s4)

21:
(fly droneb s4 s5 b1 b1 b0)

22:
(inspect droneb s5)

23:
(deliver-supply droneb water1 s5)


Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (supply-at medkit1 depot)
Deleting (empty dronea)
Adding (carrying dronea medkit1)
Updating (total-cost) (0) by 1 increase
```

```
Checking next happening (time 2)
Deleting (at dronea depot)
Deleting (battery dronea b6)
Adding (at dronea s1)
Adding (battery dronea b5)
Updating (total-cost) (1) by 1 increase

Checking next happening (time 3)
Deleting (carrying dronea medkit1)
Deleting (needs medkit1 s1)
Adding (empty dronea)
Adding (delivered medkit1 s1)
Updating (total-cost) (2) by 1 increase

Checking next happening (time 4)
Deleting (at droneb charging2)
Deleting (battery droneb b6)
Adding (at droneb s3)
Adding (battery droneb b5)
Updating (total-cost) (3) by 1 increase

Checking next happening (time 5)
Adding (inspected s3)
Updating (total-cost) (4) by 1 increase

Checking next happening (time 6)
Deleting (at dronea s1)
Deleting (battery dronea b5)
Adding (at dronea depot)
Adding (battery dronea b4)
Updating (total-cost) (5) by 1 increase

Checking next happening (time 7)
Deleting (supply-at medkit2 depot)
Deleting (empty dronea)
Adding (carrying dronea medkit2)
Updating (total-cost) (6) by 1 increase

Checking next happening (time 8)
Deleting (at dronea depot)
Deleting (battery dronea b4)
Adding (at dronea s1)
Adding (battery dronea b3)
Updating (total-cost) (7) by 1 increase

Checking next happening (time 9)
Deleting (at dronea s1)
Deleting (battery dronea b3)
Adding (at dronea s2)
Adding (battery dronea b1)
Updating (total-cost) (8) by 2 increase
```

```
Checking next happening (time 10)
Adding (inspected s2)
Updating (total-cost) (10) by 1 increase

Checking next happening (time 11)
Deleting (carrying dronea medkit2)
Deleting (needs medkit2 s2)
Adding (empty dronea)
Adding (delivered medkit2 s2)
Updating (total-cost) (11) by 1 increase

Checking next happening (time 12)
Deleting (at droneb s3)
Deleting (battery droneb b5)
Adding (at droneb depot)
Adding (battery droneb b3)
Updating (total-cost) (12) by 2 increase

Checking next happening (time 13)
Deleting (supply-at water1 depot)
Deleting (empty droneb)
Adding (carrying droneb water1)
Updating (total-cost) (14) by 1 increase

Checking next happening (time 14)
Deleting (at droneb depot)
Deleting (battery droneb b3)
Adding (at droneb s3)
Adding (battery droneb b1)
Updating (total-cost) (15) by 2 increase

Checking next happening (time 15)
Deleting (at droneb s3)
Deleting (battery droneb b1)
Adding (at droneb charging2)
Adding (battery droneb b0)
Updating (total-cost) (17) by 1 increase

Checking next happening (time 16)
Deleting (battery droneb b0)
Adding (battery droneb b1)
Updating (total-cost) (18) by 1 increase

Checking next happening (time 17)
Deleting (battery droneb b1)
Adding (battery droneb b2)
Updating (total-cost) (19) by 1 increase

Checking next happening (time 18)
Deleting (battery droneb b2)
```

```
Adding (battery droneb b3)
Updating (total-cost) (20) by 1 increase

Checking next happening (time 19)
Deleting (at droneb charging2)
Deleting (battery droneb b3)
Adding (at droneb s4)
Adding (battery droneb b1)
Updating (total-cost) (21) by 2 increase

Checking next happening (time 20)
Adding (inspected s4)
Adding (accessible s5)
Updating (total-cost) (23) by 1 increase

Checking next happening (time 21)
Deleting (at droneb s4)
Deleting (battery droneb b1)
Adding (at droneb s5)
Adding (battery droneb b0)
Updating (total-cost) (24) by 1 increase

Checking next happening (time 22)
Adding (inspected s5)
Updating (total-cost) (25) by 1 increase

Checking next happening (time 23)
Deleting (carrying droneb water1)
Deleting (needs water1 s5)
Adding (empty droneb)
Adding (delivered water1 s5)
Updating (total-cost) (26) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 27

Successful plans:
Value: 27
 plans/medium-B-lama 27
```

A4.13 small-A-astar

None

```
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
```

```
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/small-A-astar
Plan to validate:

Plan size: 11
1:
(pick-up-supply dronea kit1 depot)

2:
(fly dronea depot charging b4 b1 b3)

3:
(fly dronea charging s3 b3 b1 b2)

4:
(inspect dronea s3)

5:
(fly dronea s3 charging b2 b1 b1)

6:
(recharge dronea charging b1 b4)

7:
(fly dronea charging depot b4 b1 b3)

8:
(fly dronea depot s1 b3 b1 b2)

9:
(deliver-supply dronea kit1 s1)

10:
(fly dronea s1 s2 b2 b1 b1)

11:
(inspect dronea s2)


Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (supply-at kit1 depot)
Deleting (empty dronea)
Adding (carrying dronea kit1)
Updating (total-cost) (0) by 1 increase
```

```
Checking next happening (time 2)
Deleting (at dronea depot)
Deleting (battery dronea b4)
Adding (at dronea charging)
Adding (battery dronea b3)
Updating (total-cost) (1) by 1 increase

Checking next happening (time 3)
Deleting (at dronea charging)
Deleting (battery dronea b3)
Adding (at dronea s3)
Adding (battery dronea b2)
Updating (total-cost) (2) by 1 increase

Checking next happening (time 4)
Adding (inspected s3)
Updating (total-cost) (3) by 1 increase

Checking next happening (time 5)
Deleting (at dronea s3)
Deleting (battery dronea b2)
Adding (at dronea charging)
Adding (battery dronea b1)
Updating (total-cost) (4) by 1 increase

Checking next happening (time 6)
Deleting (battery dronea b1)
Adding (battery dronea b4)
Updating (total-cost) (5) by 1 increase

Checking next happening (time 7)
Deleting (at dronea charging)
Deleting (battery dronea b4)
Adding (at dronea depot)
Adding (battery dronea b3)
Updating (total-cost) (6) by 1 increase

Checking next happening (time 8)
Deleting (at dronea depot)
Deleting (battery dronea b3)
Adding (at dronea s1)
Adding (battery dronea b2)
Updating (total-cost) (7) by 1 increase

Checking next happening (time 9)
Deleting (carrying dronea kit1)
Deleting (needs kit1 s1)
Adding (empty dronea)
Adding (delivered kit1 s1)
Updating (total-cost) (8) by 1 increase
```

```
Checking next happening (time 10)
Deleting (at dronea s1)
Deleting (battery dronea b2)
Adding (at dronea s2)
Adding (battery dronea b1)
Updating (total-cost) (9) by 1 increase

Checking next happening (time 11)
Adding (inspected s2)
Updating (total-cost) (10) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 11

Successful plans:
Value: 11
 plans/small-A-astar 11
```

A4.14 small-A-lama

None

```
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/small-A-lama
Plan to validate:

Plan size: 11
1:
(pick-up-supply dronea kit1 depot)

2:
(fly dronea depot charging b4 b1 b3)

3:
(fly dronea charging s3 b3 b1 b2)

4:
(inspect dronea s3)

5:
```

```
(fly dronea s3 charging b2 b1 b1)

6:
(recharge dronea charging b1 b4)

7:
(fly dronea charging depot b4 b1 b3)

8:
(fly dronea depot s1 b3 b1 b2)

9:
(deliver-supply dronea kit1 s1)

10:
(fly dronea s1 s2 b2 b1 b1)

11:
(inspect dronea s2)


Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (supply-at kit1 depot)
Deleting (empty dronea)
Adding (carrying dronea kit1)
Updating (total-cost) (0) by 1 increase

Checking next happening (time 2)
Deleting (at dronea depot)
Deleting (battery dronea b4)
Adding (at dronea charging)
Adding (battery dronea b3)
Updating (total-cost) (1) by 1 increase

Checking next happening (time 3)
Deleting (at dronea charging)
Deleting (battery dronea b3)
Adding (at dronea s3)
Adding (battery dronea b2)
Updating (total-cost) (2) by 1 increase

Checking next happening (time 4)
Adding (inspected s3)
Updating (total-cost) (3) by 1 increase

Checking next happening (time 5)
Deleting (at dronea s3)
Deleting (battery dronea b2)
```

```
Adding (at dronea charging)
Adding (battery dronea b1)
Updating (total-cost) (4) by 1 increase

Checking next happening (time 6)
Deleting (battery dronea b1)
Adding (battery dronea b4)
Updating (total-cost) (5) by 1 increase

Checking next happening (time 7)
Deleting (at dronea charging)
Deleting (battery dronea b4)
Adding (at dronea depot)
Adding (battery dronea b3)
Updating (total-cost) (6) by 1 increase

Checking next happening (time 8)
Deleting (at dronea depot)
Deleting (battery dronea b3)
Adding (at dronea s1)
Adding (battery dronea b2)
Updating (total-cost) (7) by 1 increase

Checking next happening (time 9)
Deleting (carrying dronea kit1)
Deleting (needs kit1 s1)
Adding (empty dronea)
Adding (delivered kit1 s1)
Updating (total-cost) (8) by 1 increase

Checking next happening (time 10)
Deleting (at dronea s1)
Deleting (battery dronea b2)
Adding (at dronea s2)
Adding (battery dronea b1)
Updating (total-cost) (9) by 1 increase

Checking next happening (time 11)
Adding (inspected s2)
Updating (total-cost) (10) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 11

Successful plans:
Value: 11
 plans/small-A-lama 11
```

A4.15 small-B-astar

None

```
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/small-B-astar
Plan to validate:

Plan size: 12
1:
(pick-up-supply dronea kit1 depot)

2:
(fly dronea depot charging b4 b1 b3)

3:
(fly dronea charging s3 b3 b1 b2)

4:
(inspect dronea s3)

5:
(fly dronea s3 charging b2 b1 b1)

6:
(recharge dronea charging b1 b2)

7:
(recharge dronea charging b2 b3)

8:
(fly dronea charging depot b3 b1 b2)

9:
(fly dronea depot s1 b2 b1 b1)

10:
(deliver-supply dronea kit1 s1)

11:
(fly dronea s1 s2 b1 b1 b0)

12:
(inspect dronea s2)
```

```
Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (supply-at kit1 depot)
Deleting (empty dronea)
Adding (carrying dronea kit1)
Updating (total-cost) (0) by 1 increase

Checking next happening (time 2)
Deleting (at dronea depot)
Deleting (battery dronea b4)
Adding (at dronea charging)
Adding (battery dronea b3)
Updating (total-cost) (1) by 1 increase

Checking next happening (time 3)
Deleting (at dronea charging)
Deleting (battery dronea b3)
Adding (at dronea s3)
Adding (battery dronea b2)
Updating (total-cost) (2) by 1 increase

Checking next happening (time 4)
Adding (inspected s3)
Updating (total-cost) (3) by 1 increase

Checking next happening (time 5)
Deleting (at dronea s3)
Deleting (battery dronea b2)
Adding (at dronea charging)
Adding (battery dronea b1)
Updating (total-cost) (4) by 1 increase

Checking next happening (time 6)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (5) by 1 increase

Checking next happening (time 7)
Deleting (battery dronea b2)
Adding (battery dronea b3)
Updating (total-cost) (6) by 1 increase

Checking next happening (time 8)
Deleting (at dronea charging)
Deleting (battery dronea b3)
Adding (at dronea depot)
Adding (battery dronea b2)
Updating (total-cost) (7) by 1 increase
```

```
Checking next happening (time 9)
Deleting (at dronea depot)
Deleting (battery dronea b2)
Adding (at dronea s1)
Adding (battery dronea b1)
Updating (total-cost) (8) by 1 increase

Checking next happening (time 10)
Deleting (carrying dronea kit1)
Deleting (needs kit1 s1)
Adding (empty dronea)
Adding (delivered kit1 s1)
Updating (total-cost) (9) by 1 increase

Checking next happening (time 11)
Deleting (at dronea s1)
Deleting (battery dronea b1)
Adding (at dronea s2)
Adding (battery dronea b0)
Updating (total-cost) (10) by 1 increase

Checking next happening (time 12)
Adding (inspected s2)
Updating (total-cost) (11) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 12

Successful plans:
Value: 12
 plans/small-B-astar 12
```

A4.16 small-B-lama

None

```
Type-checking fly
...action passes type checking.
Type-checking inspect
...action passes type checking.
Type-checking pick-up-supply
...action passes type checking.
Type-checking deliver-supply
...action passes type checking.
Type-checking recharge
...action passes type checking.
Checking plan: plans/small-B-lama
Plan to validate:
```

```
Plan size: 16
1:
(pick-up-supply dronea kit1 depot)

2:
(fly dronea depot s1 b4 b1 b3)

3:
(deliver-supply dronea kit1 s1)

4:
(fly dronea s1 depot b3 b1 b2)

5:
(fly dronea depot charging b2 b1 b1)

6:
(recharge dronea charging b1 b2)

7:
(fly dronea charging s3 b2 b1 b1)

8:
(inspect dronea s3)

9:
(fly dronea s3 charging b1 b1 b0)

10:
(recharge dronea charging b0 b1)

11:
(recharge dronea charging b1 b2)

12:
(recharge dronea charging b2 b3)

13:
(fly dronea charging depot b3 b1 b2)

14:
(fly dronea depot s1 b2 b1 b1)

15:
(fly dronea s1 s2 b1 b1 b0)

16:
(inspect dronea s2)
```

```
Plan Validation details
-----------------------

Checking next happening (time 1)
Deleting (supply-at kit1 depot)
Deleting (empty dronea)
Adding (carrying dronea kit1)
Updating (total-cost) (0) by 1 increase

Checking next happening (time 2)
Deleting (at dronea depot)
Deleting (battery dronea b4)
Adding (at dronea s1)
Adding (battery dronea b3)
Updating (total-cost) (1) by 1 increase

Checking next happening (time 3)
Deleting (carrying dronea kit1)
Deleting (needs kit1 s1)
Adding (empty dronea)
Adding (delivered kit1 s1)
Updating (total-cost) (2) by 1 increase

Checking next happening (time 4)
Deleting (at dronea s1)
Deleting (battery dronea b3)
Adding (at dronea depot)
Adding (battery dronea b2)
Updating (total-cost) (3) by 1 increase

Checking next happening (time 5)
Deleting (at dronea depot)
Deleting (battery dronea b2)
Adding (at dronea charging)
Adding (battery dronea b1)
Updating (total-cost) (4) by 1 increase

Checking next happening (time 6)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (5) by 1 increase

Checking next happening (time 7)
Deleting (at dronea charging)
Deleting (battery dronea b2)
Adding (at dronea s3)
Adding (battery dronea b1)
Updating (total-cost) (6) by 1 increase

Checking next happening (time 8)
Adding (inspected s3)
```

```
Updating (total-cost) (7) by 1 increase

Checking next happening (time 9)
Deleting (at dronea s3)
Deleting (battery dronea b1)
Adding (at dronea charging)
Adding (battery dronea b0)
Updating (total-cost) (8) by 1 increase

Checking next happening (time 10)
Deleting (battery dronea b0)
Adding (battery dronea b1)
Updating (total-cost) (9) by 1 increase

Checking next happening (time 11)
Deleting (battery dronea b1)
Adding (battery dronea b2)
Updating (total-cost) (10) by 1 increase

Checking next happening (time 12)
Deleting (battery dronea b2)
Adding (battery dronea b3)
Updating (total-cost) (11) by 1 increase

Checking next happening (time 13)
Deleting (at dronea charging)
Deleting (battery dronea b3)
Adding (at dronea depot)
Adding (battery dronea b2)
Updating (total-cost) (12) by 1 increase

Checking next happening (time 14)
Deleting (at dronea depot)
Deleting (battery dronea b2)
Adding (at dronea s1)
Adding (battery dronea b1)
Updating (total-cost) (13) by 1 increase

Checking next happening (time 15)
Deleting (at dronea s1)
Deleting (battery dronea b1)
Adding (at dronea s2)
Adding (battery dronea b0)
Updating (total-cost) (14) by 1 increase

Checking next happening (time 16)
Adding (inspected s2)
Updating (total-cost) (15) by 1 increase
Plan executed successfully - checking goal
Plan valid
Final value: 16
```

```
Successful plans:
Value: 16
 plans/small-B-lama 16
```